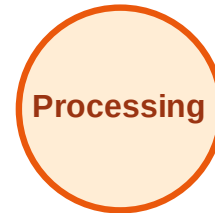


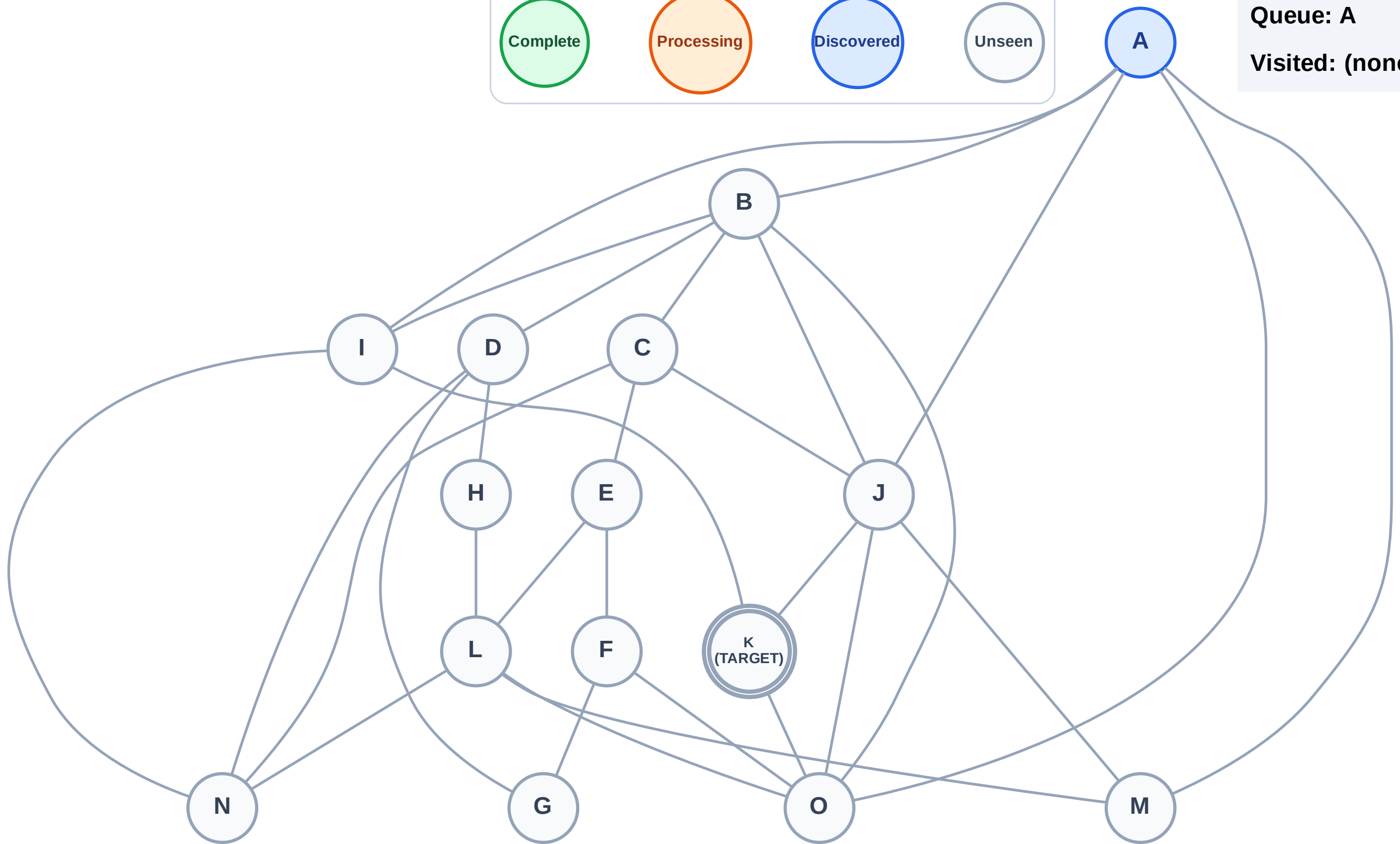
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

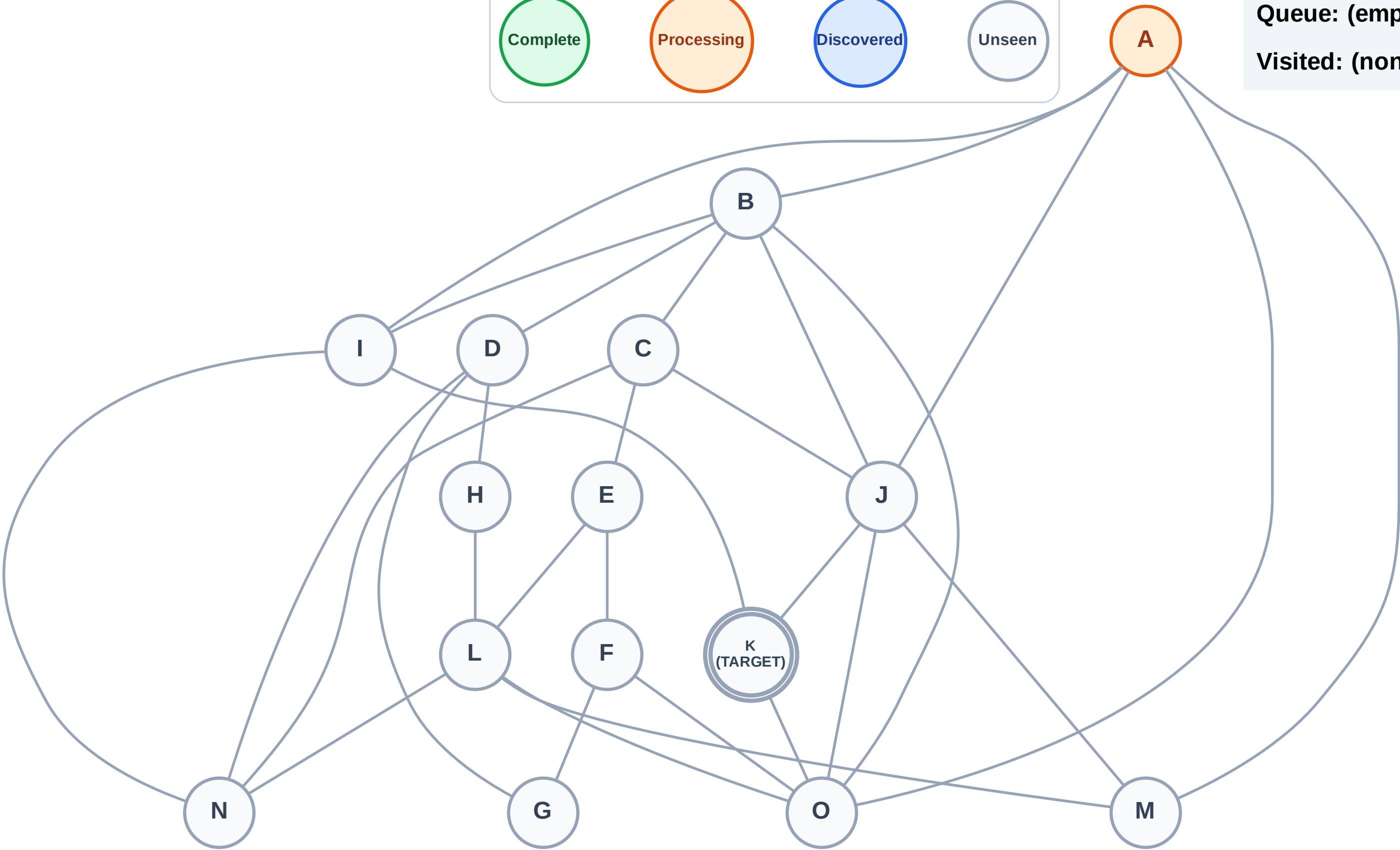
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



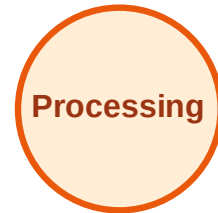
BFS Traversal

Step 3: Discover B, I, J, M, O from A

Legend



Complete



Processing



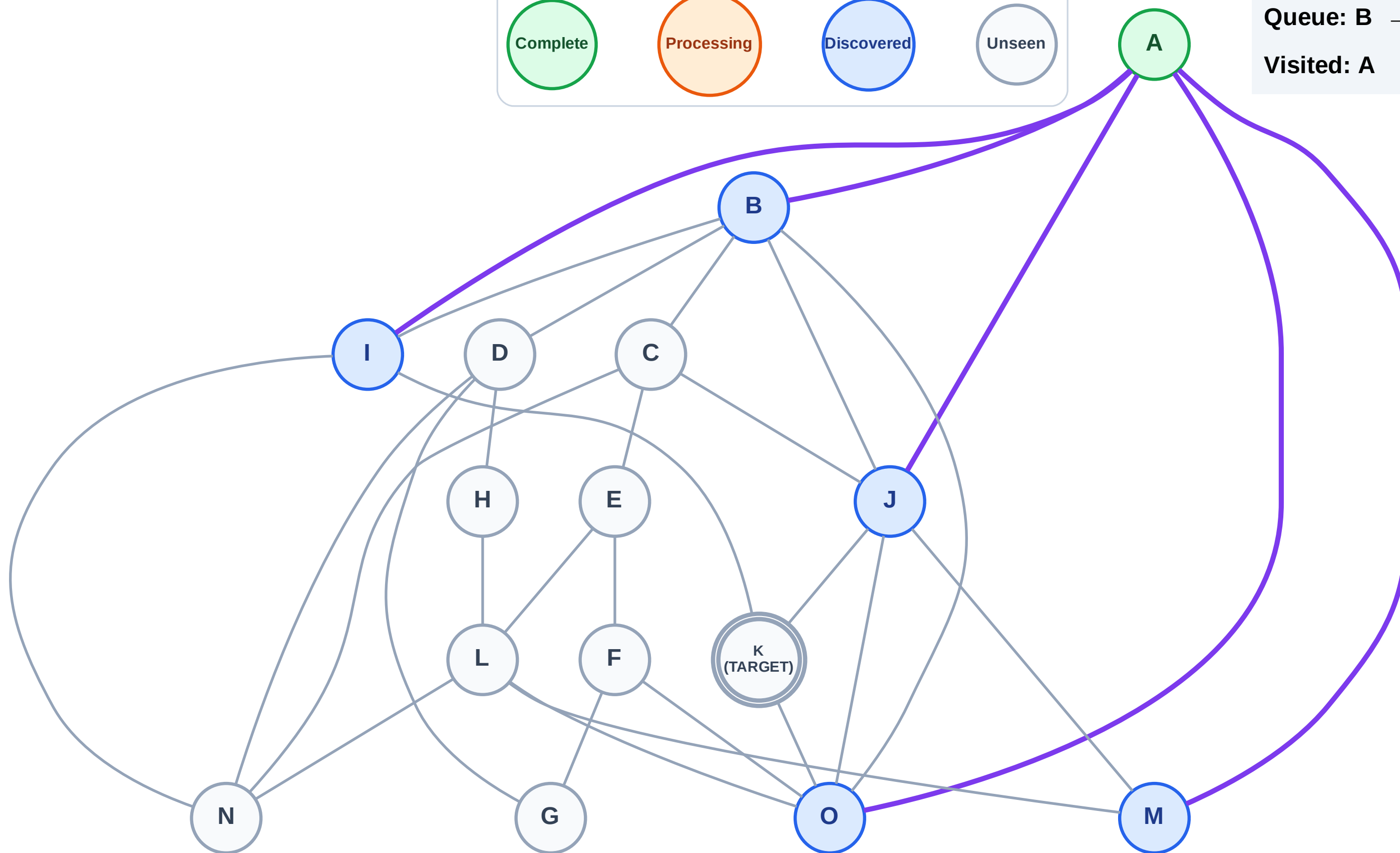
Discovered



Unseen

Queue: B → I → J → M → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

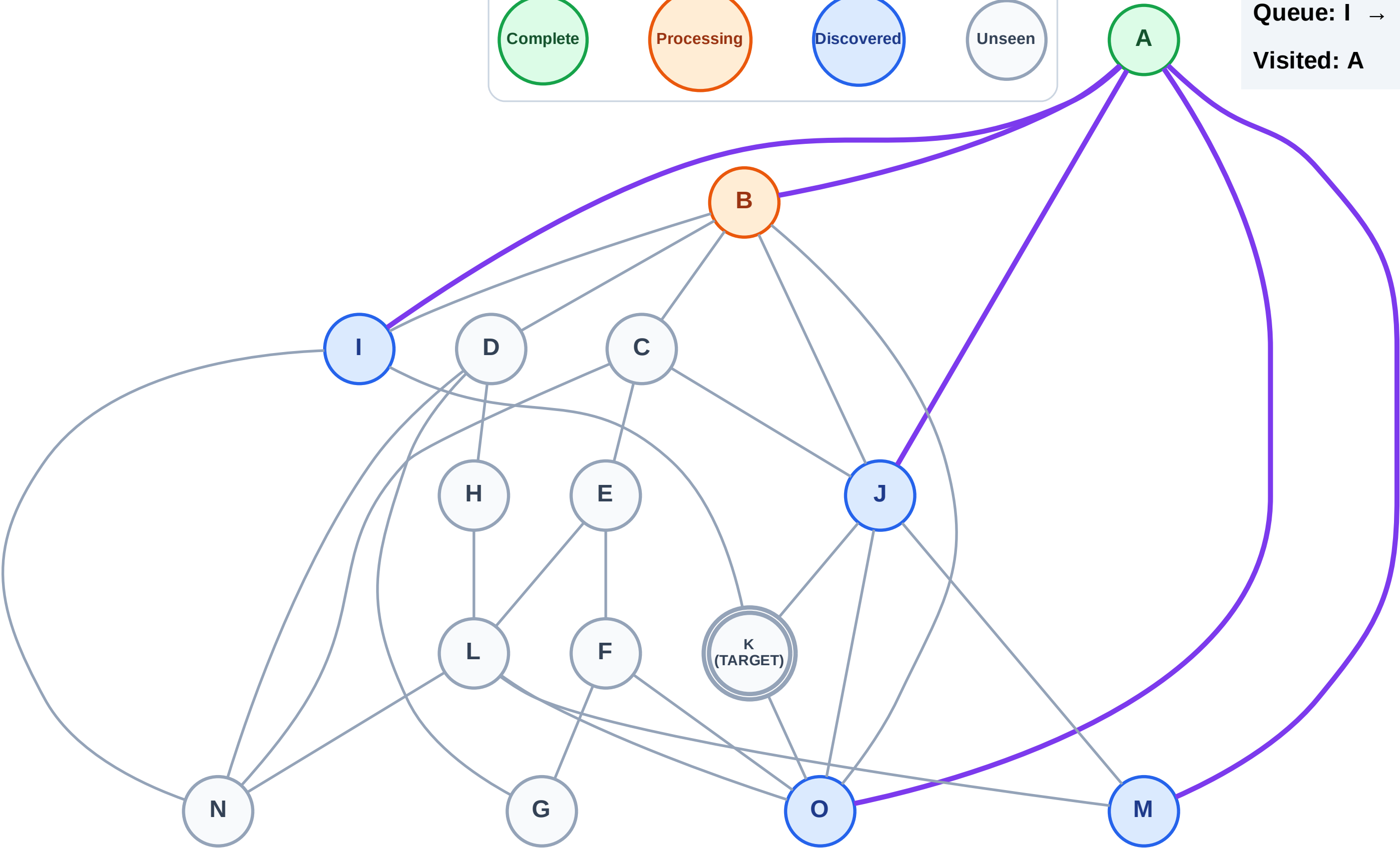
Processing

Discovered

Unseen

Queue: I → J → M → O

Visited: A



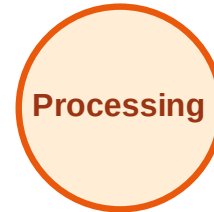
BFS Traversal

Step 5: Discover C, D from B

Legend



Complete



Processing

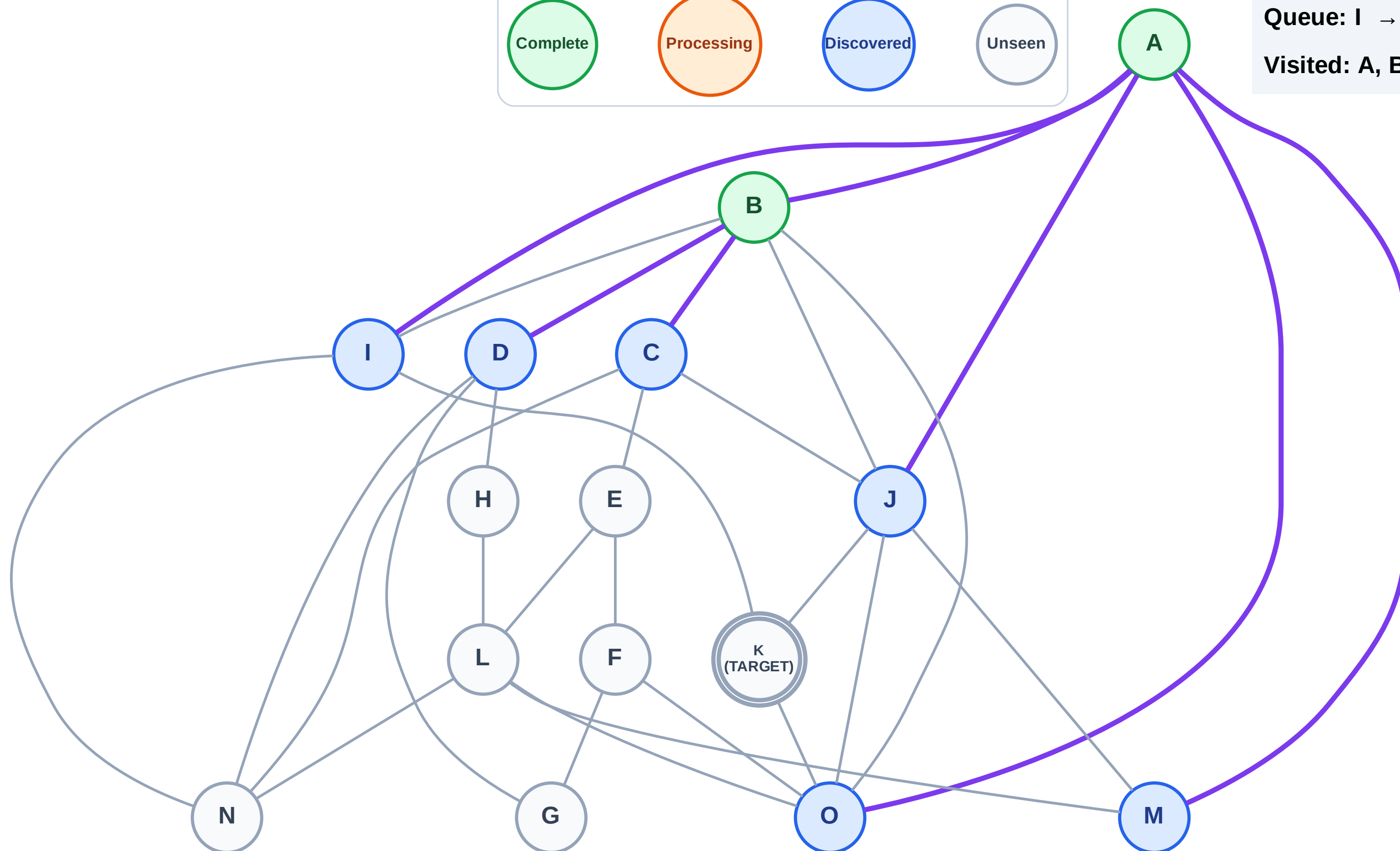


Discovered



Unseen

Queue: I → J → M → O → C → D
Visited: A, B



BFS Traversal

Step 6: Process node I

Legend

Complete

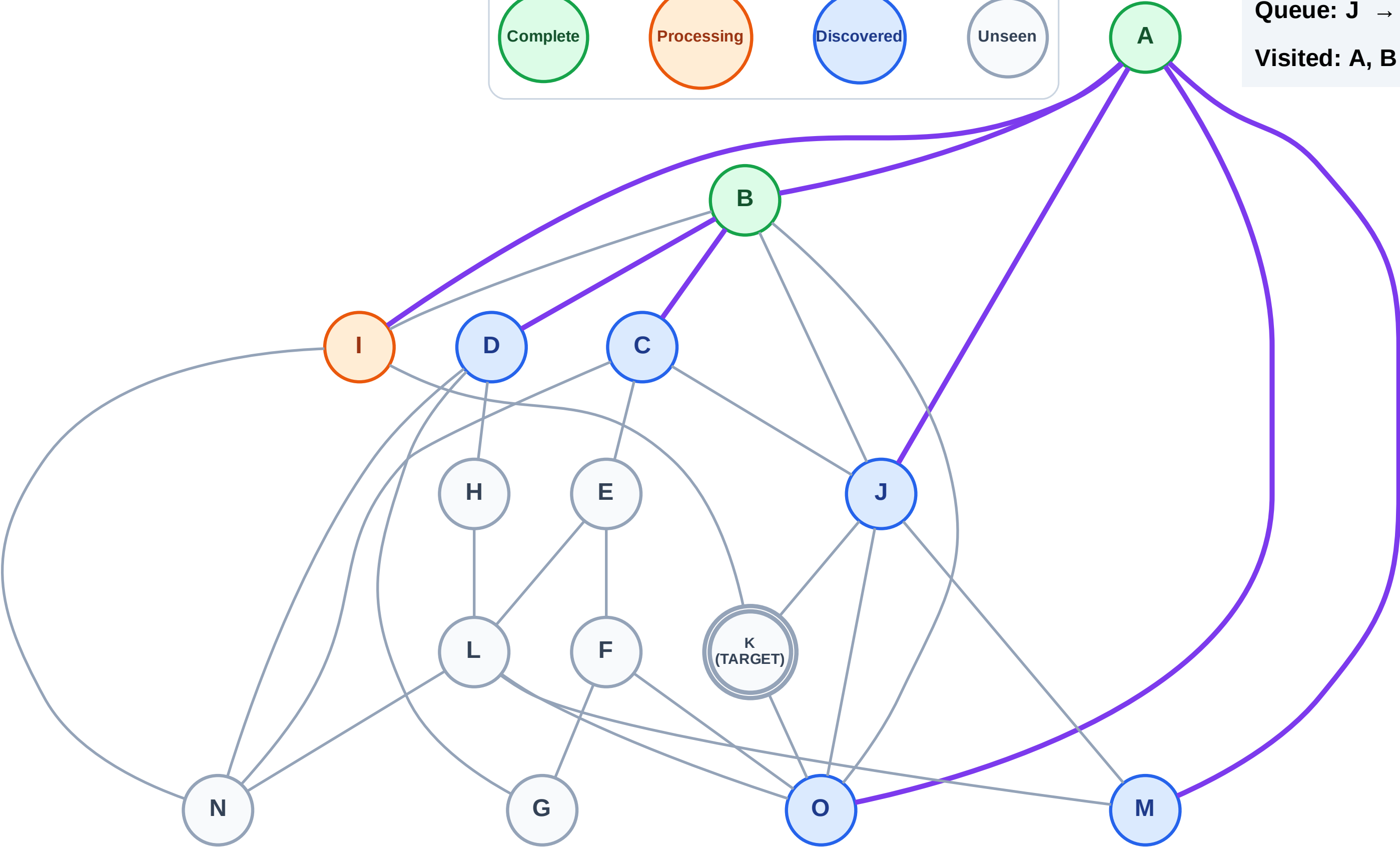
Processing

Discovered

Unseen

Queue: J → M → O → C → D

Visited: A, B



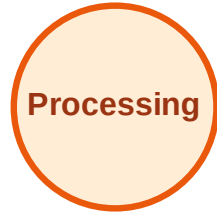
BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → I → K

Legend



Complete



Processing



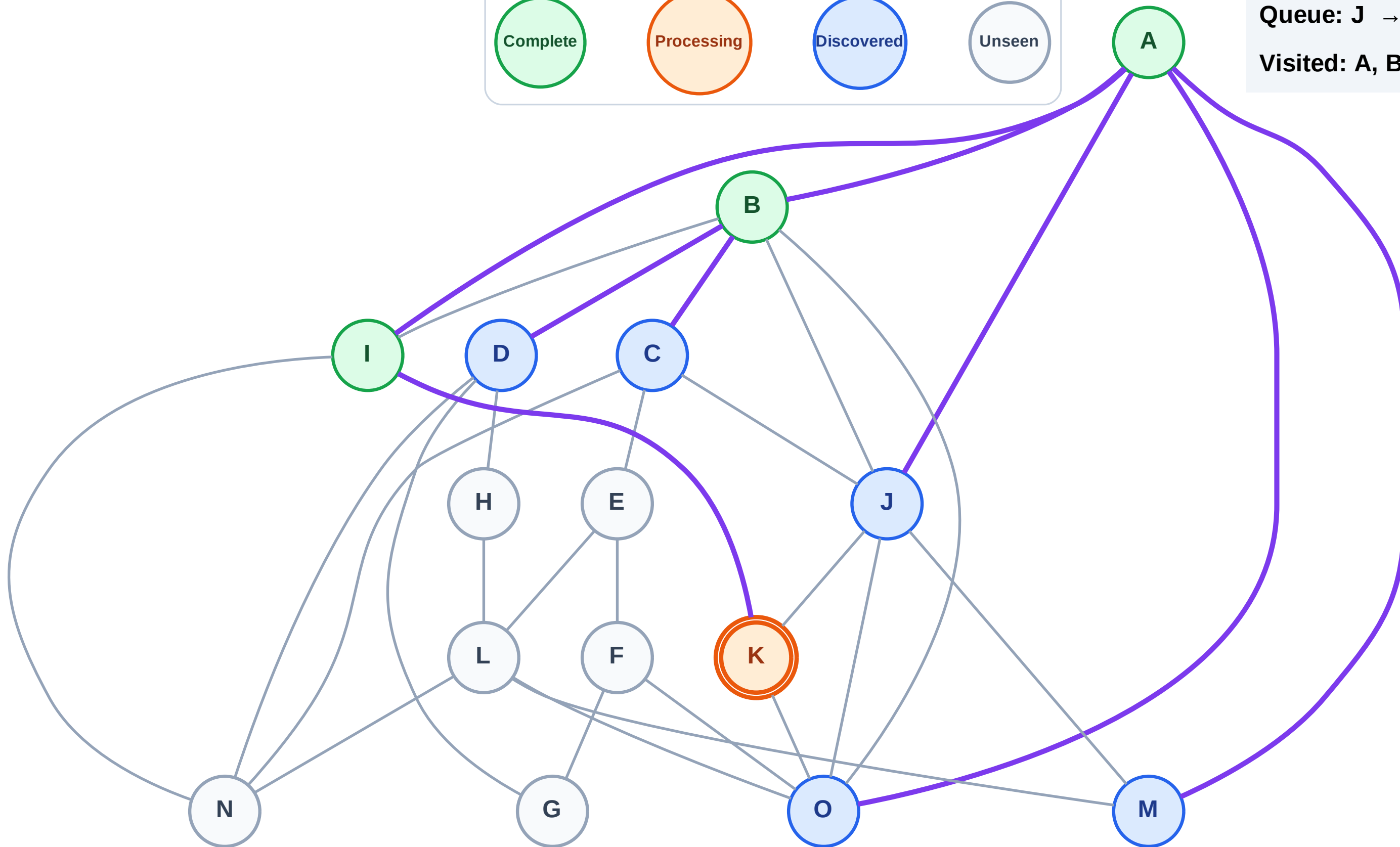
Discovered



Unseen

Queue: J → M → O → C → D

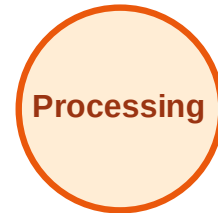
Visited: A, B, I, K



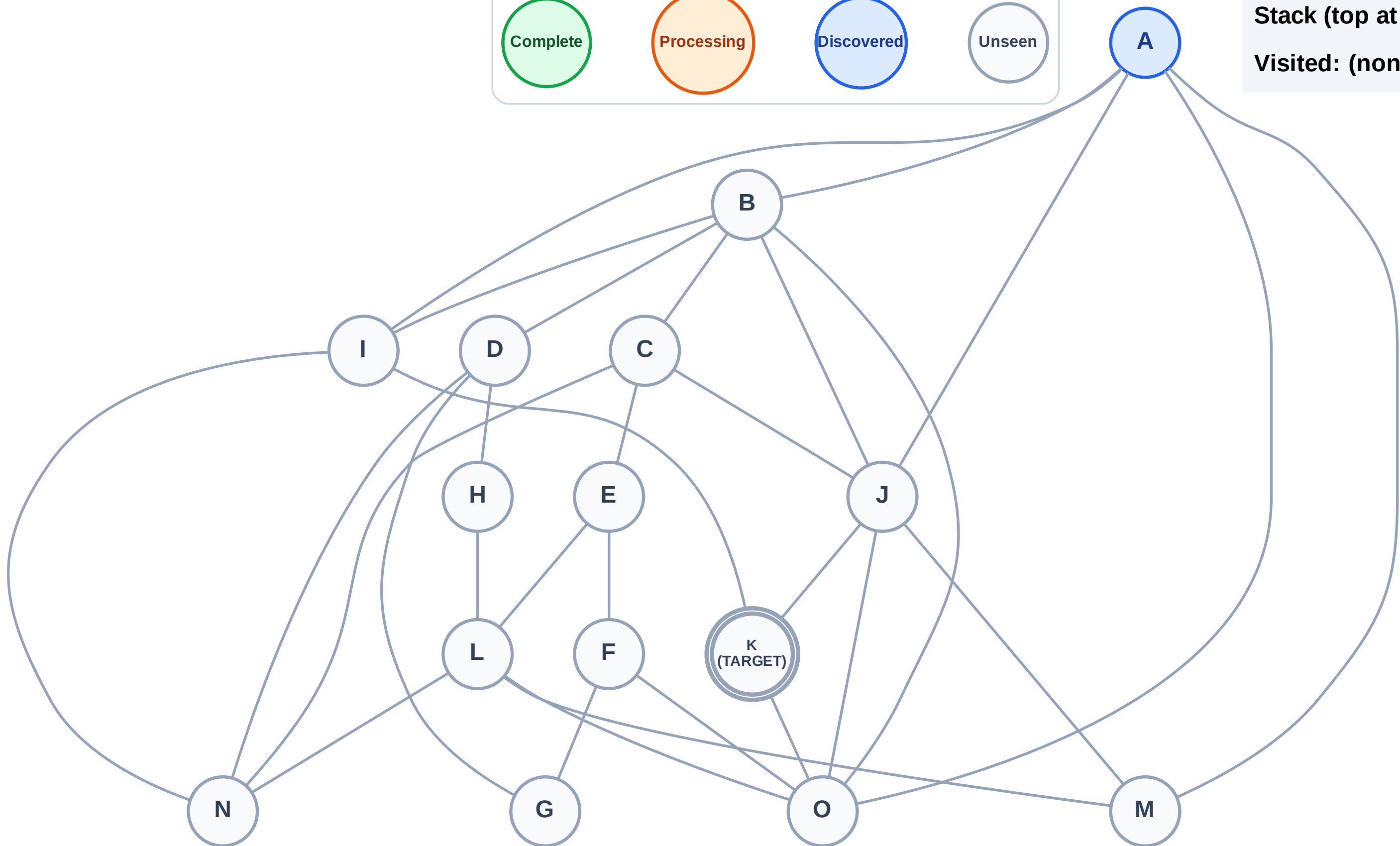
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

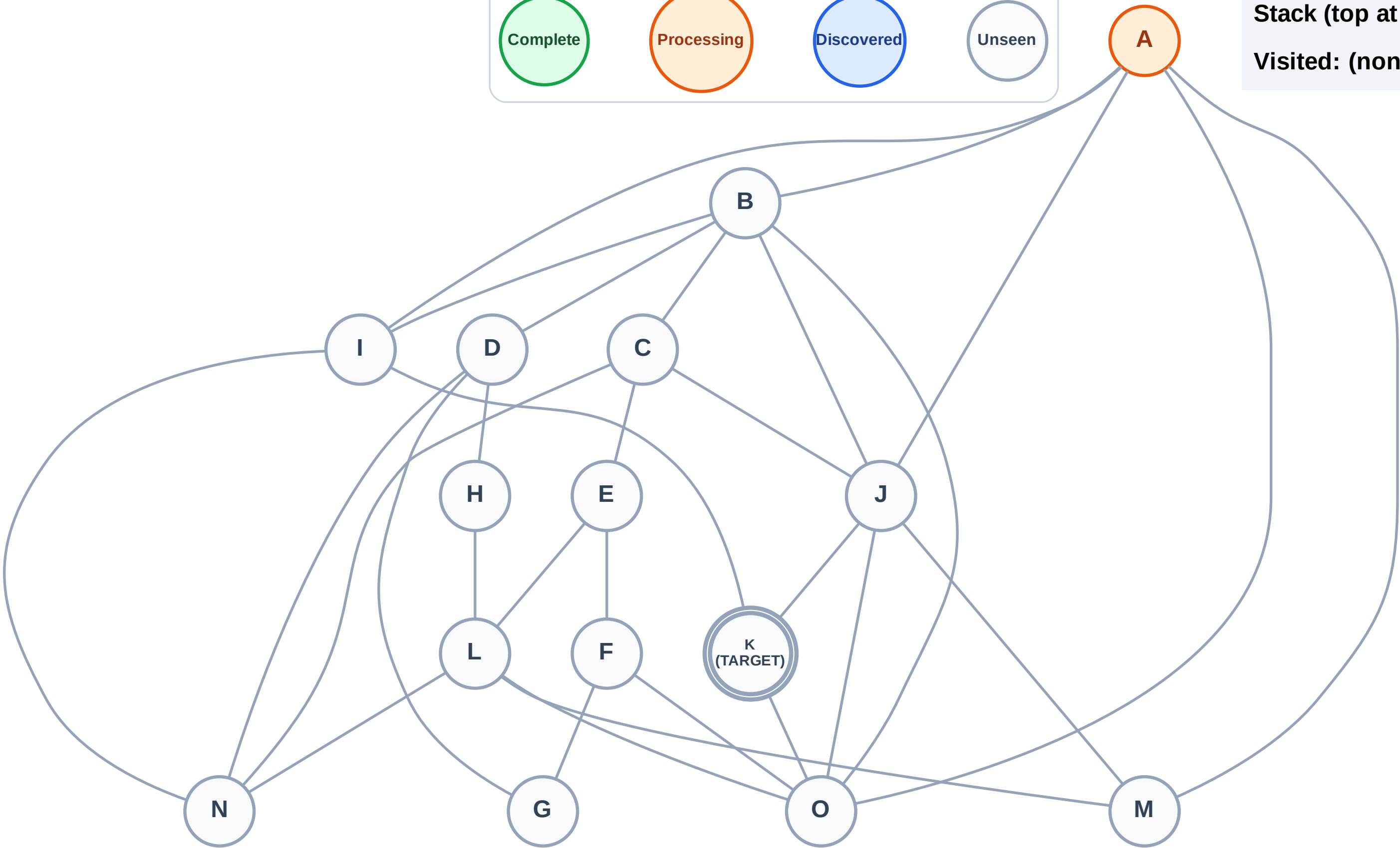
Complete

Processing

Discovered

Unseen

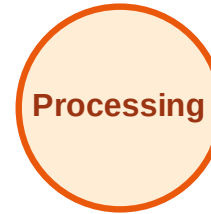
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

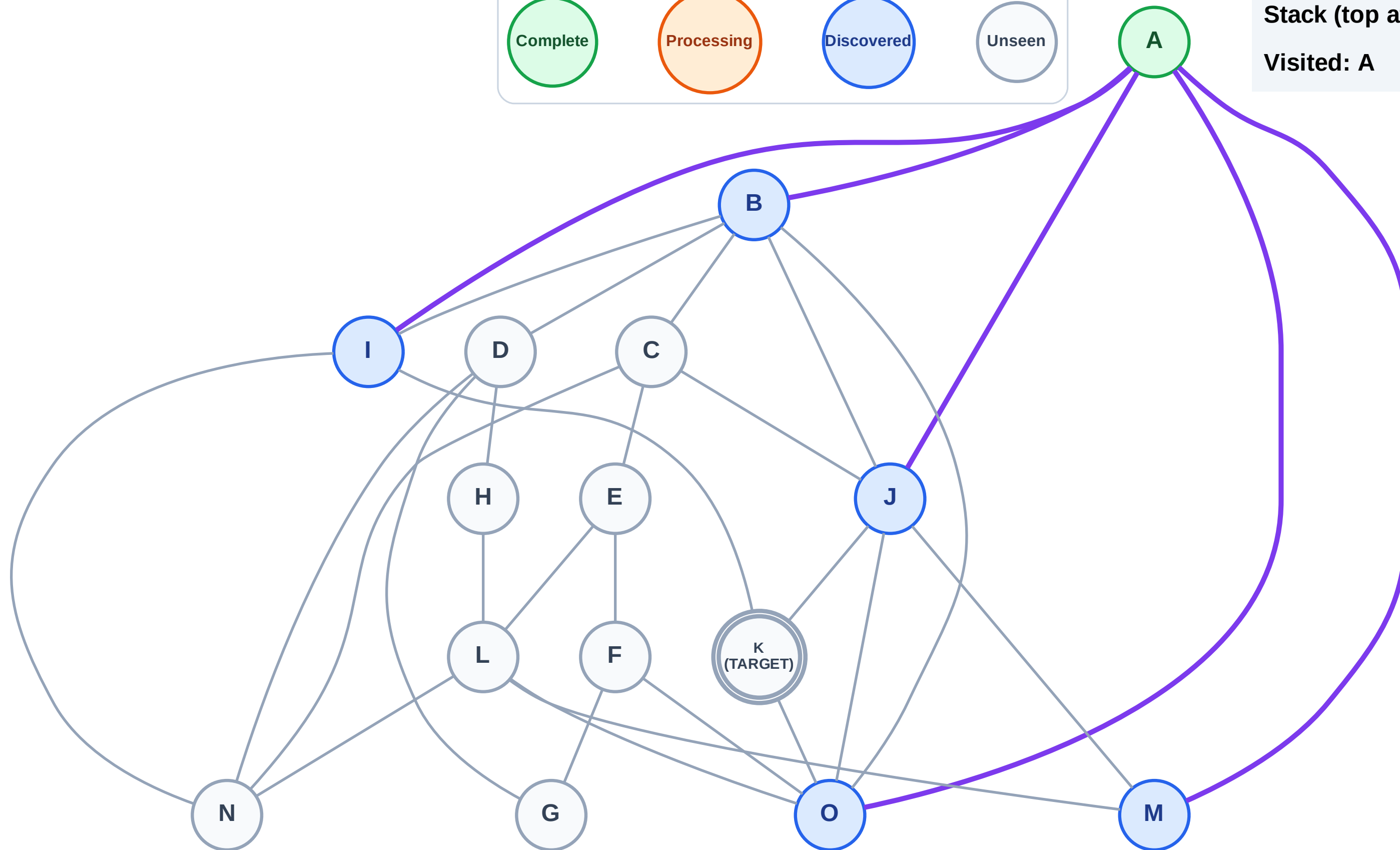
Step 3: Discover O, M, J, I, B from A

Legend



Stack (top at right): O | M | J | I | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

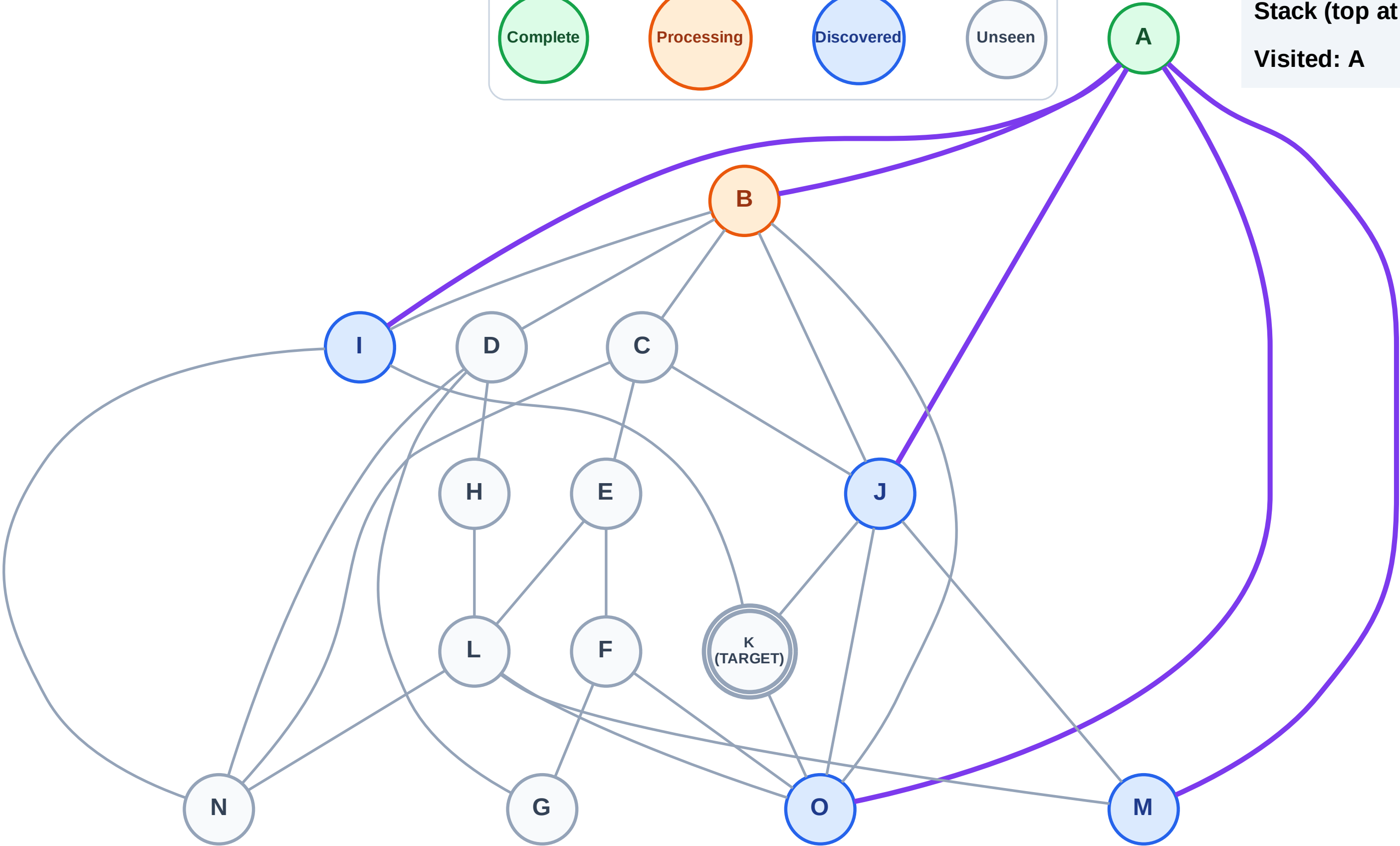
Processing

Discovered

Unseen

Stack (top at right): O | M | J | I

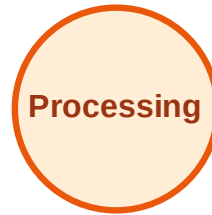
Visited: A



DFS Traversal

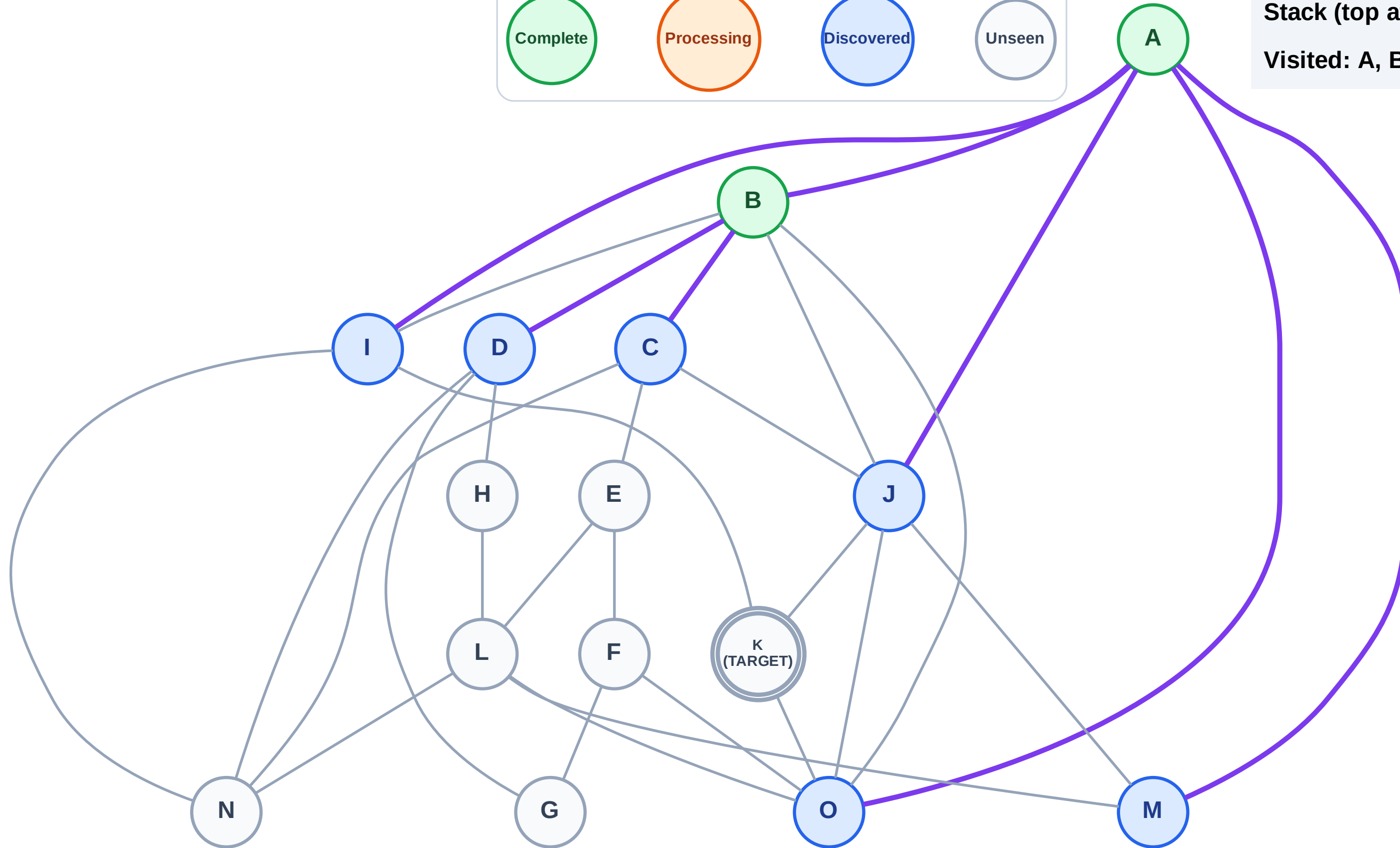
Step 5: Discover D, C from B

Legend



Stack (top at right): O | M | J | I | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

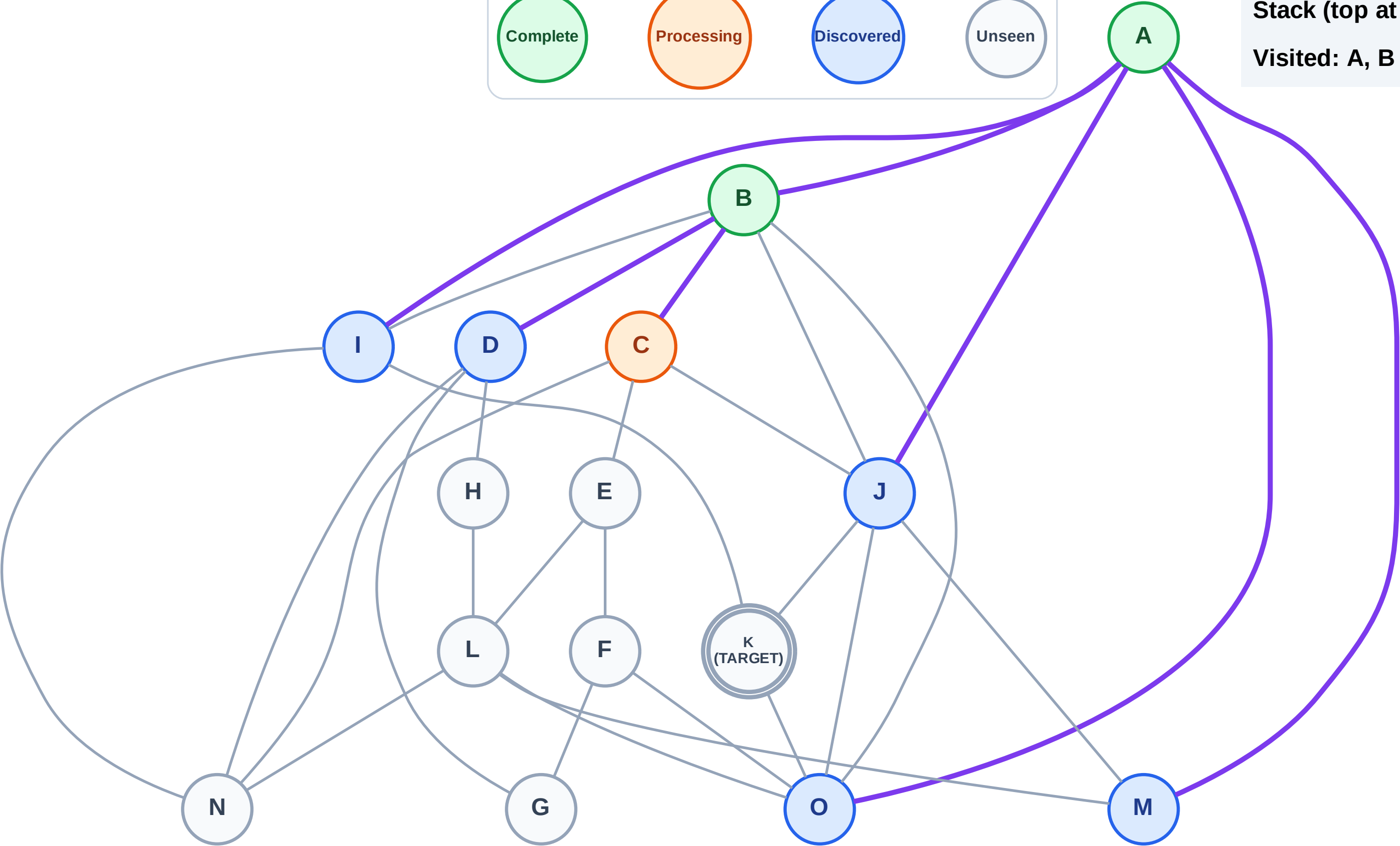
Processing

Discovered

Unseen

Stack (top at right): O | M | J | I | D

Visited: A, B



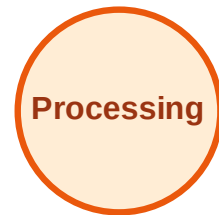
DFS Traversal

Step 7: Discover N, E from C

Legend



Complete



Processing



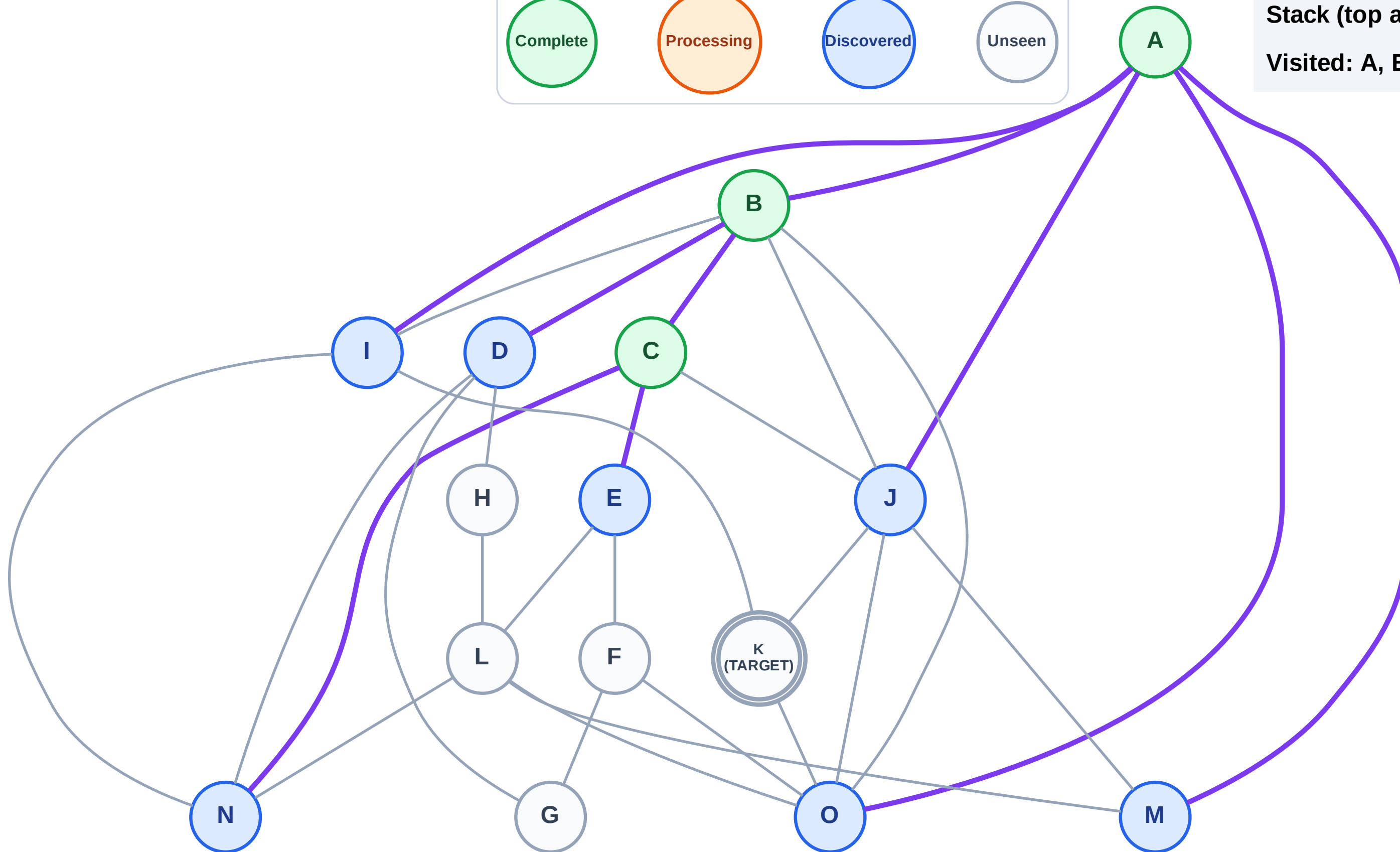
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N | E

Visited: A, B, C



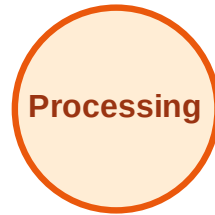
DFS Traversal

Step 8: Process node E

Legend



Complete



Processing



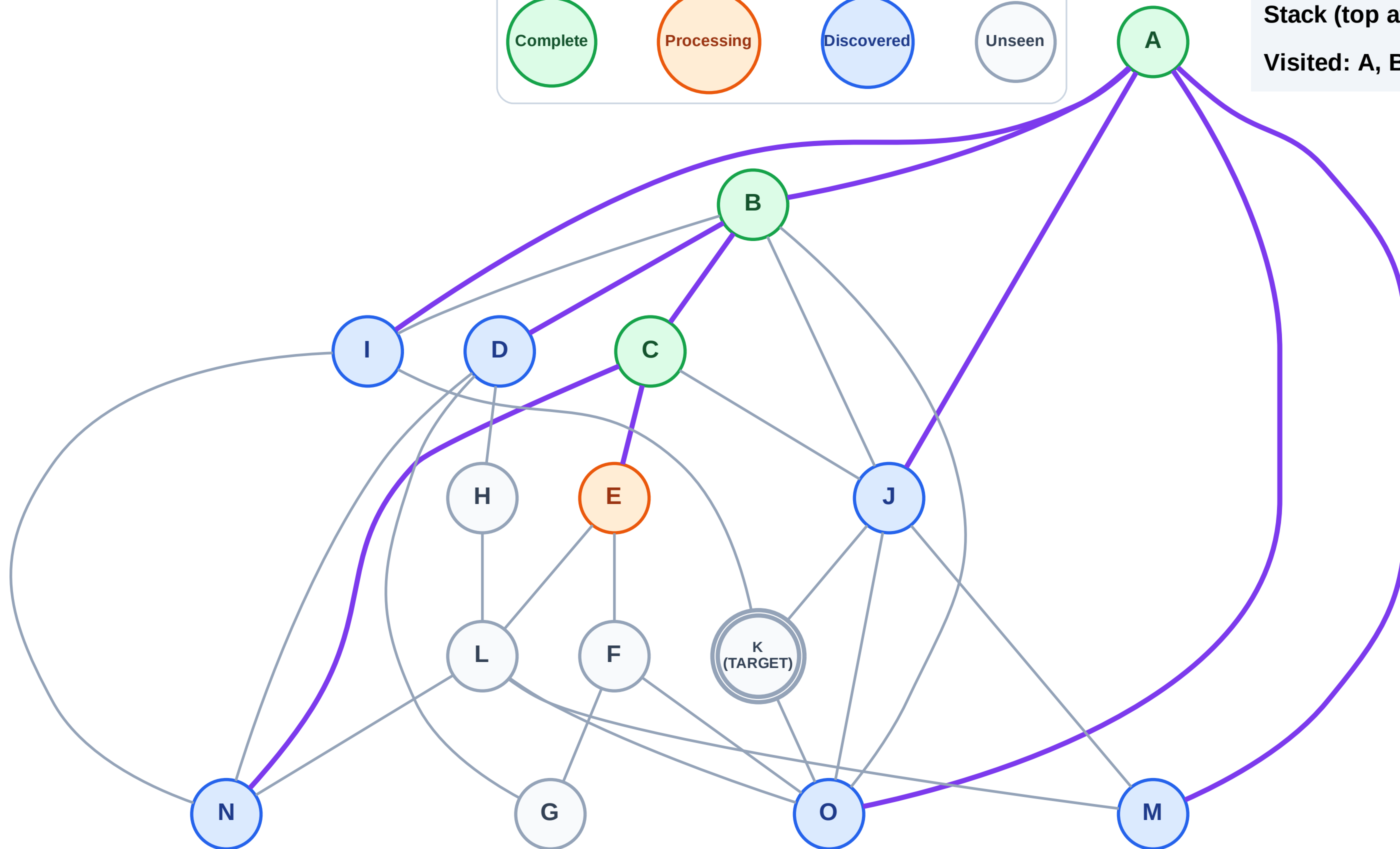
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N

Visited: A, B, C



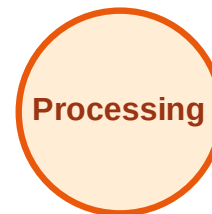
DFS Traversal

Step 9: Discover L, F from E

Legend



Complete



Processing



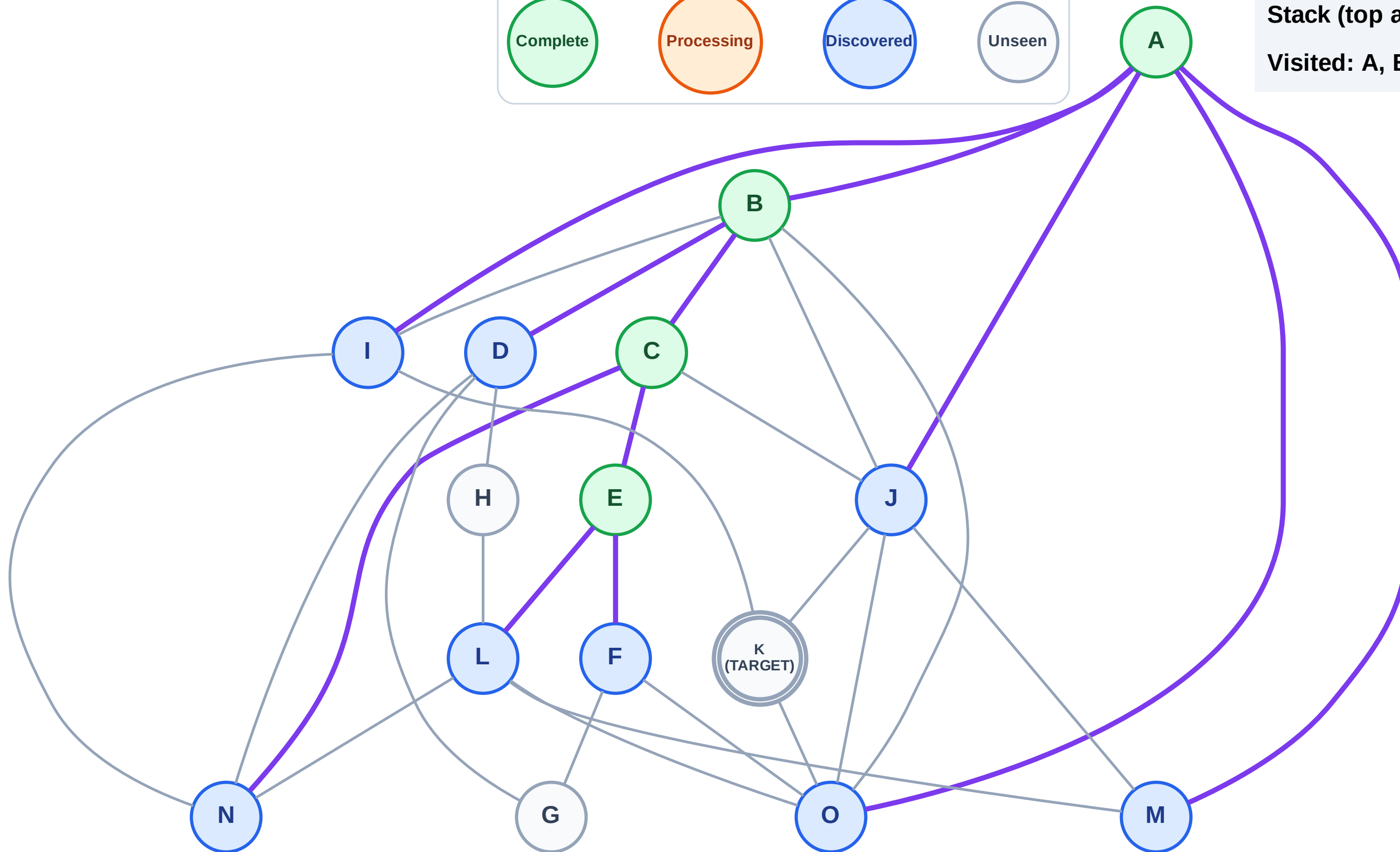
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N | L | F

Visited: A, B, C, E



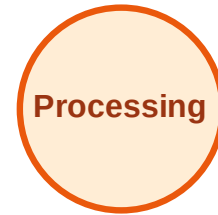
DFS Traversal

Step 10: Process node F

Legend



Complete



Processing



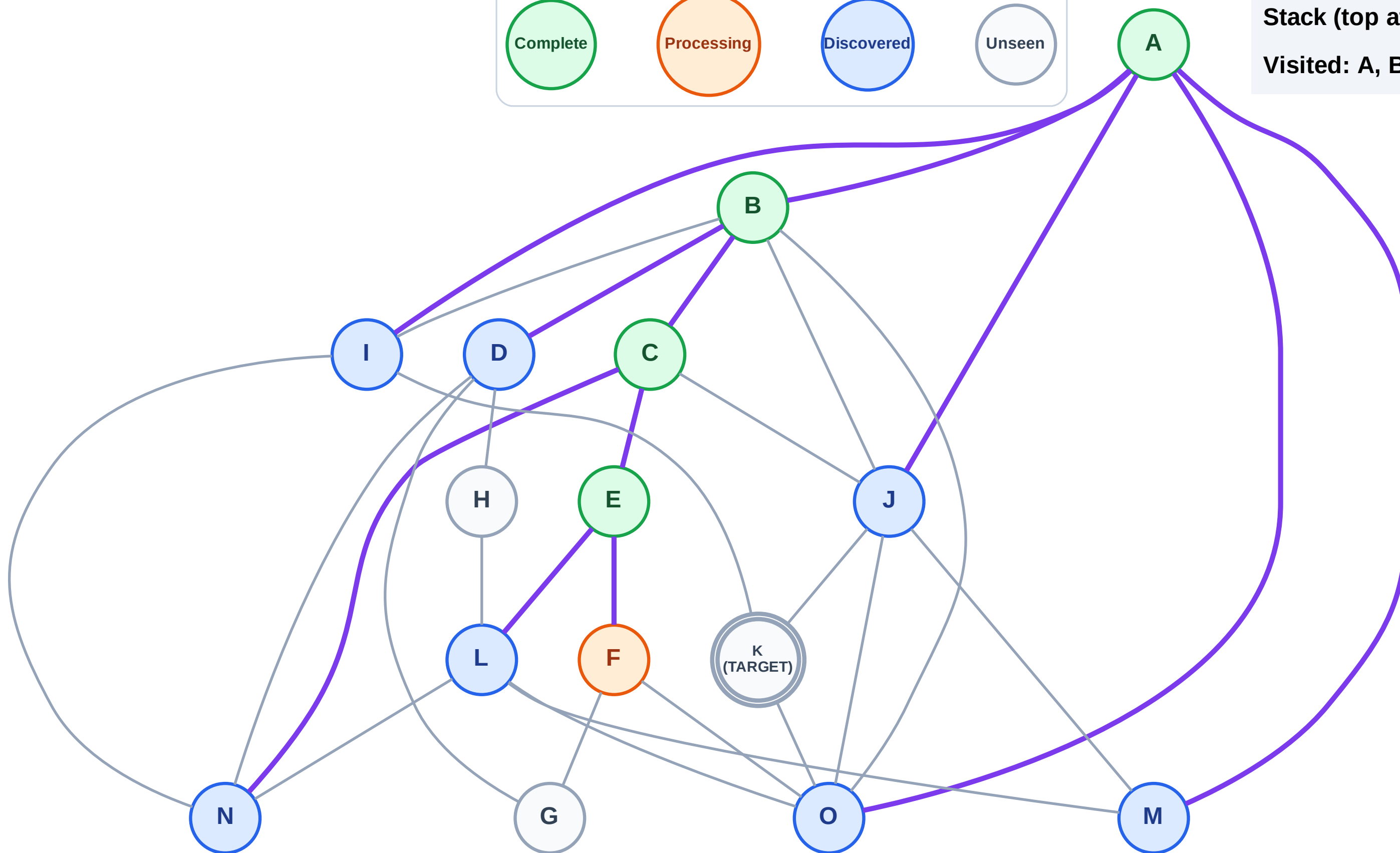
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N | L

Visited: A, B, C, E



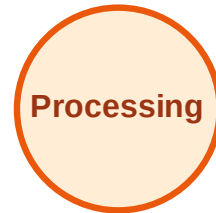
DFS Traversal

Step 11: Discover G from F

Legend



Complete



Processing



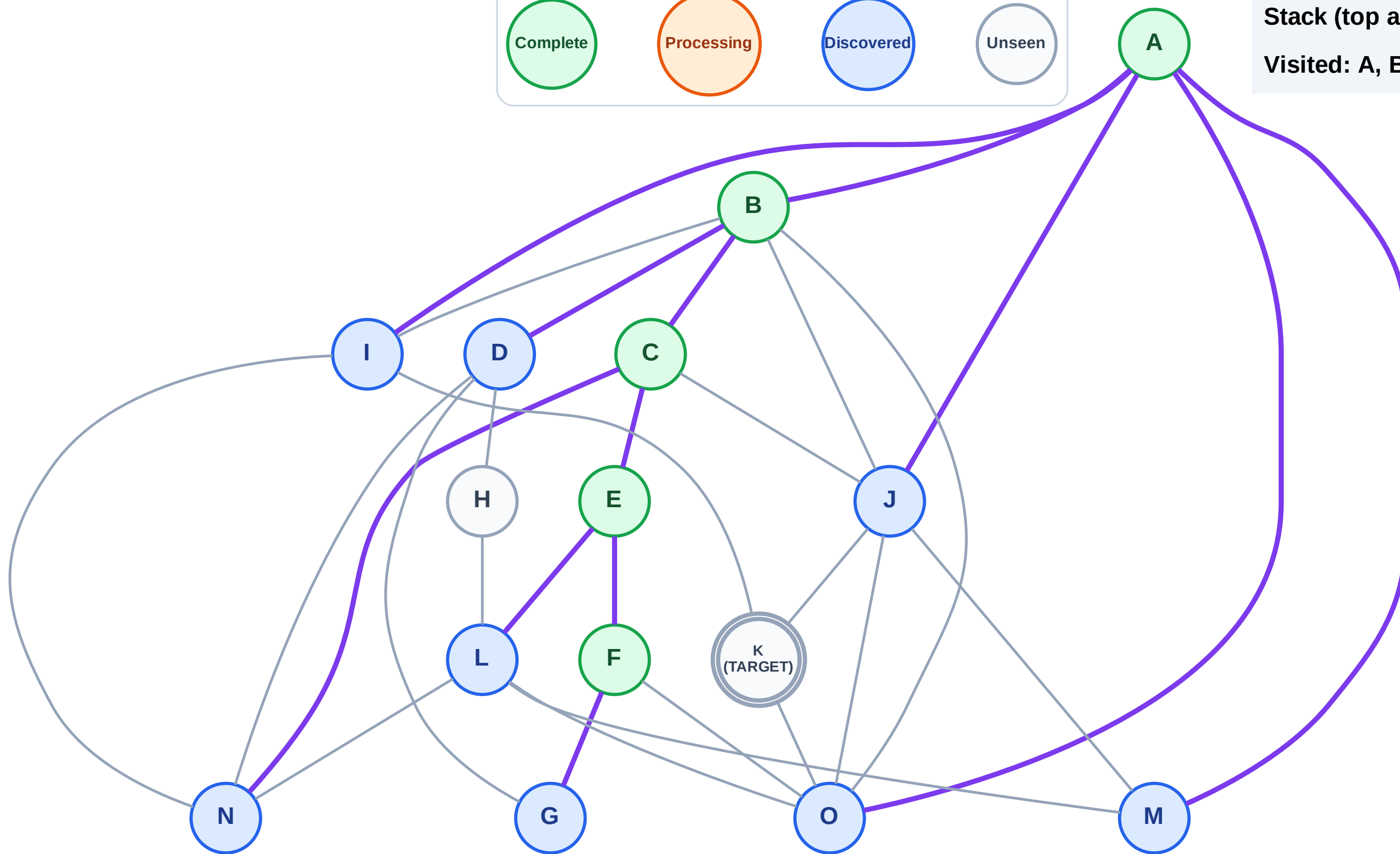
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N | L | G

Visited: A, B, C, E, F



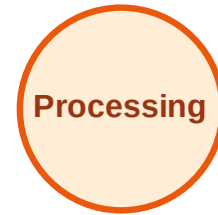
DFS Traversal

Step 12: Process node G

Legend



Complete



Processing



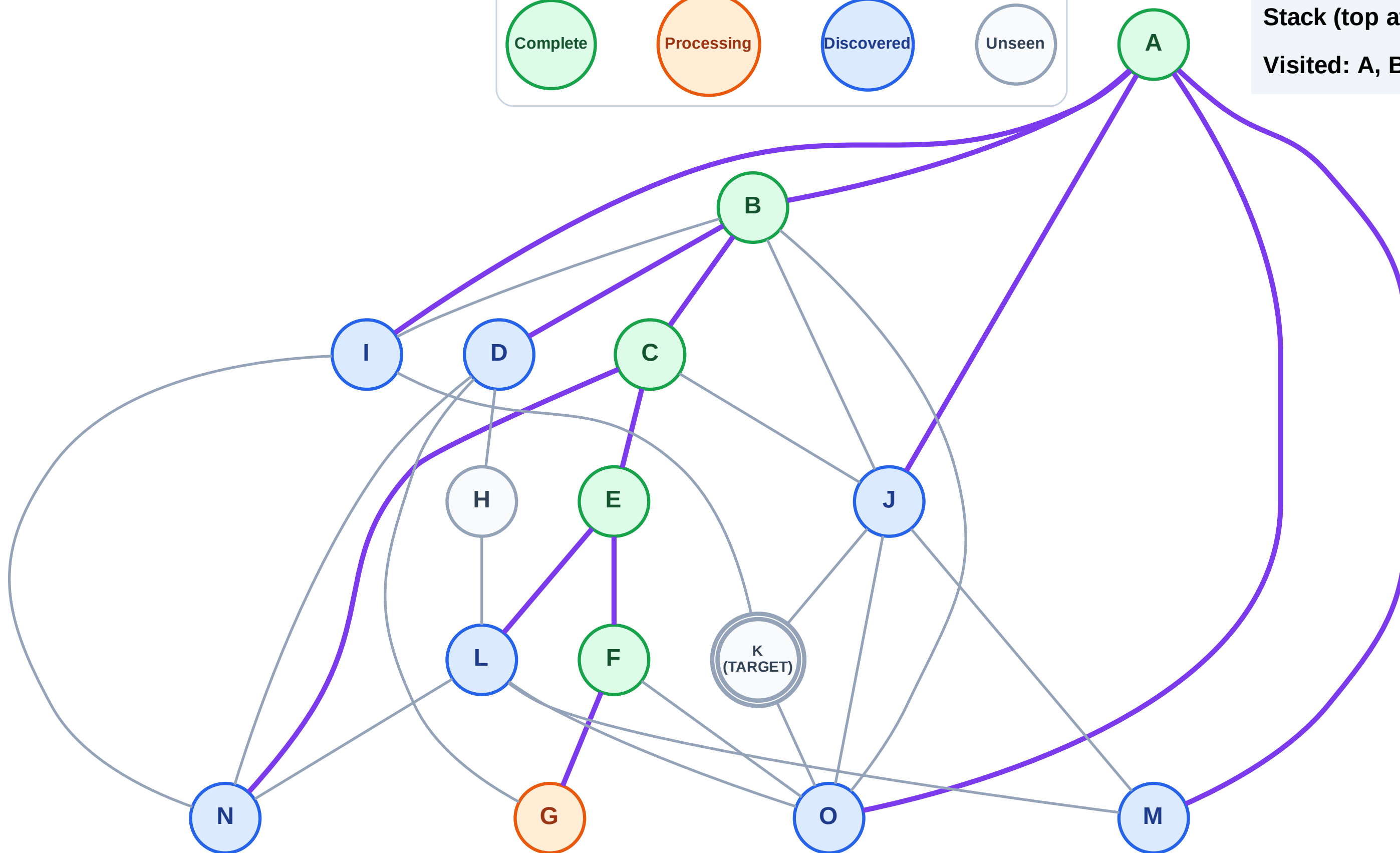
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N | L

Visited: A, B, C, E, F



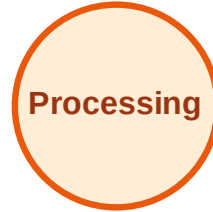
DFS Traversal

Step 13: No new neighbors from G

Legend



Complete



Processing



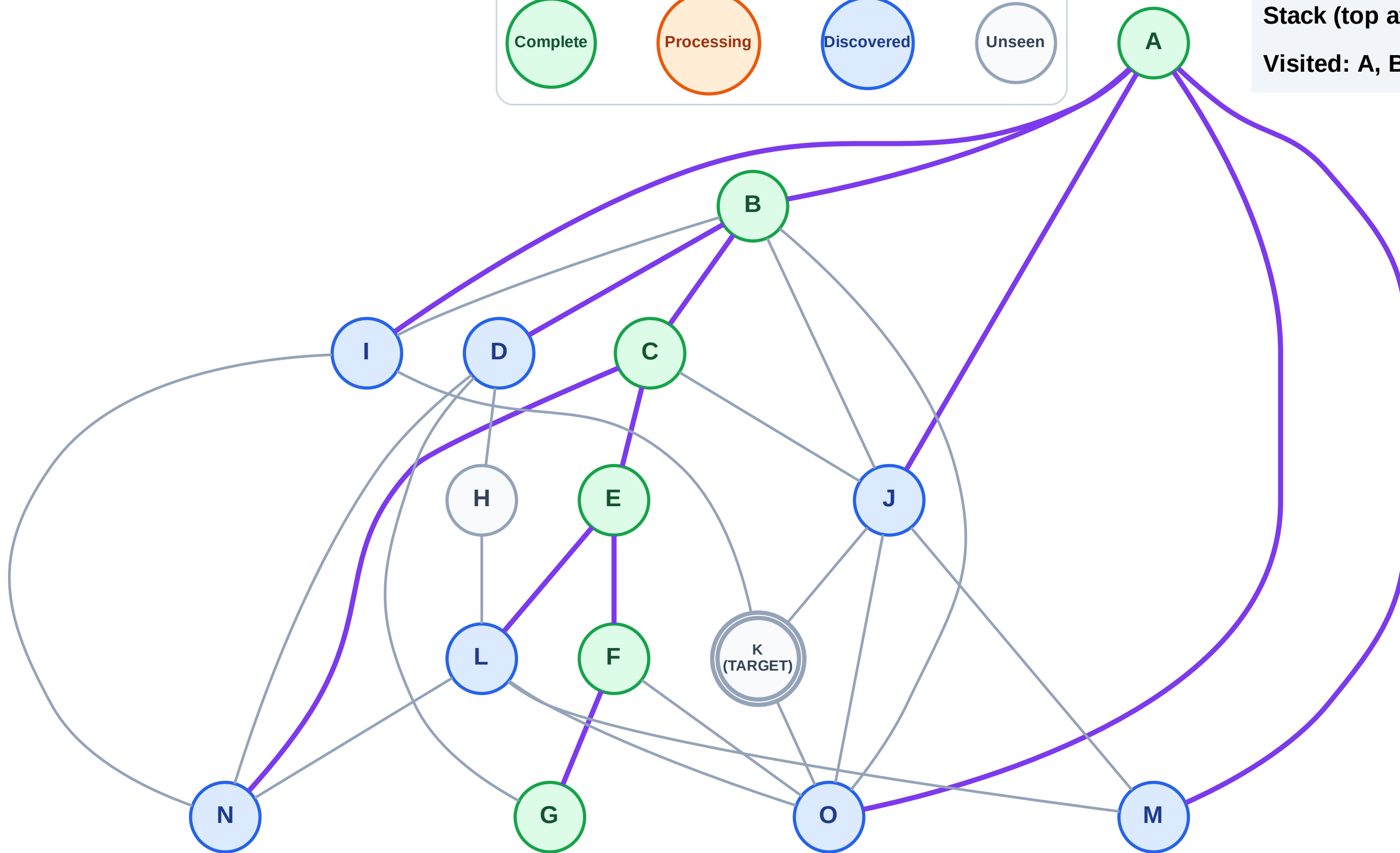
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N | L

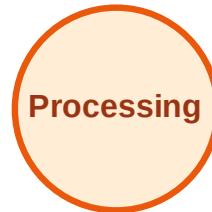
Visited: A, B, C, E, F, G



DFS Traversal

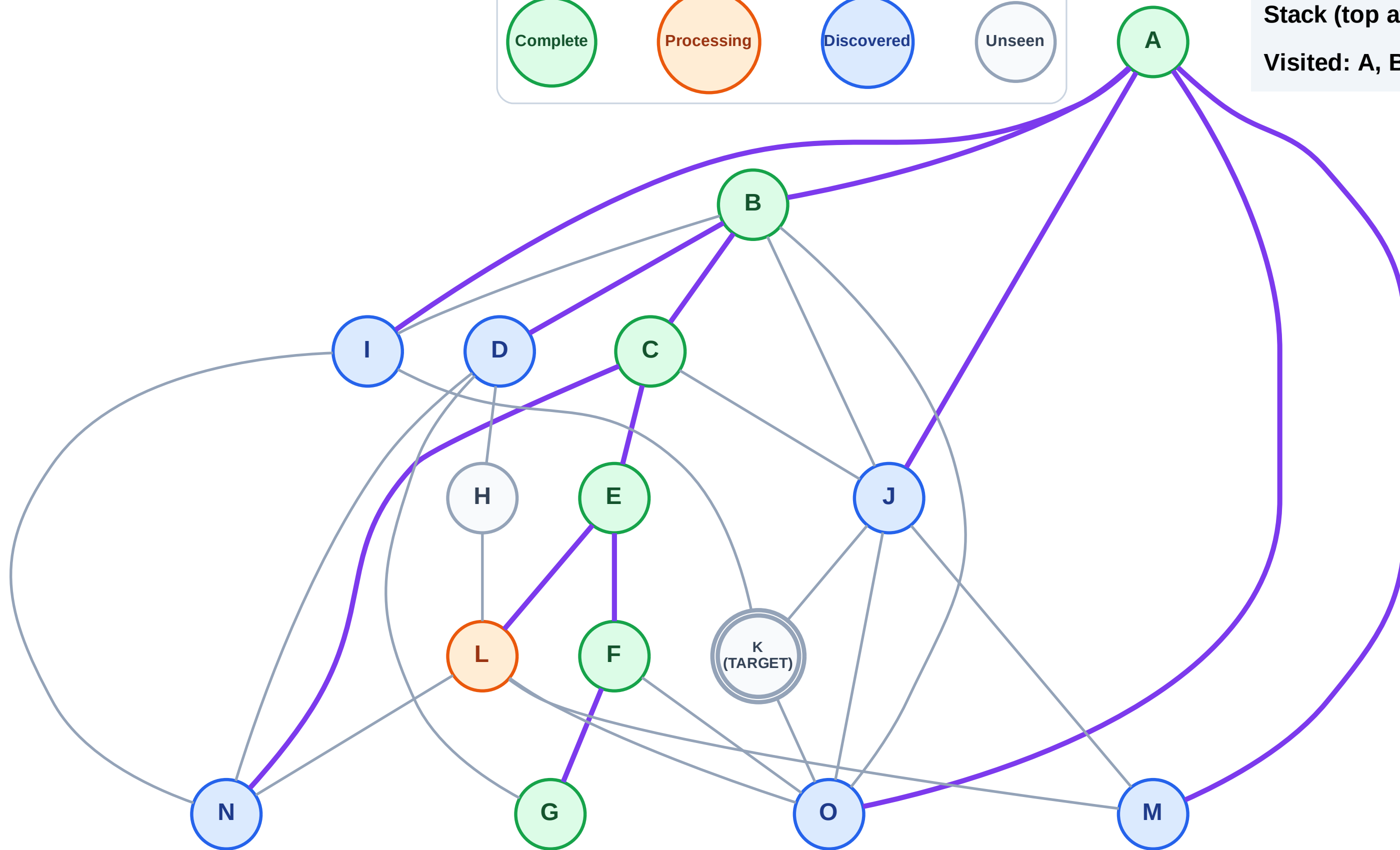
Step 14: Process node L

Legend



Stack (top at right): O | M | J | I | D | N

Visited: A, B, C, E, F, G



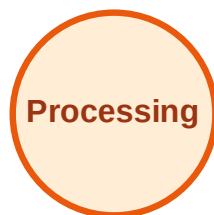
DFS Traversal

Step 15: Discover H from L

Legend



Complete



Processing



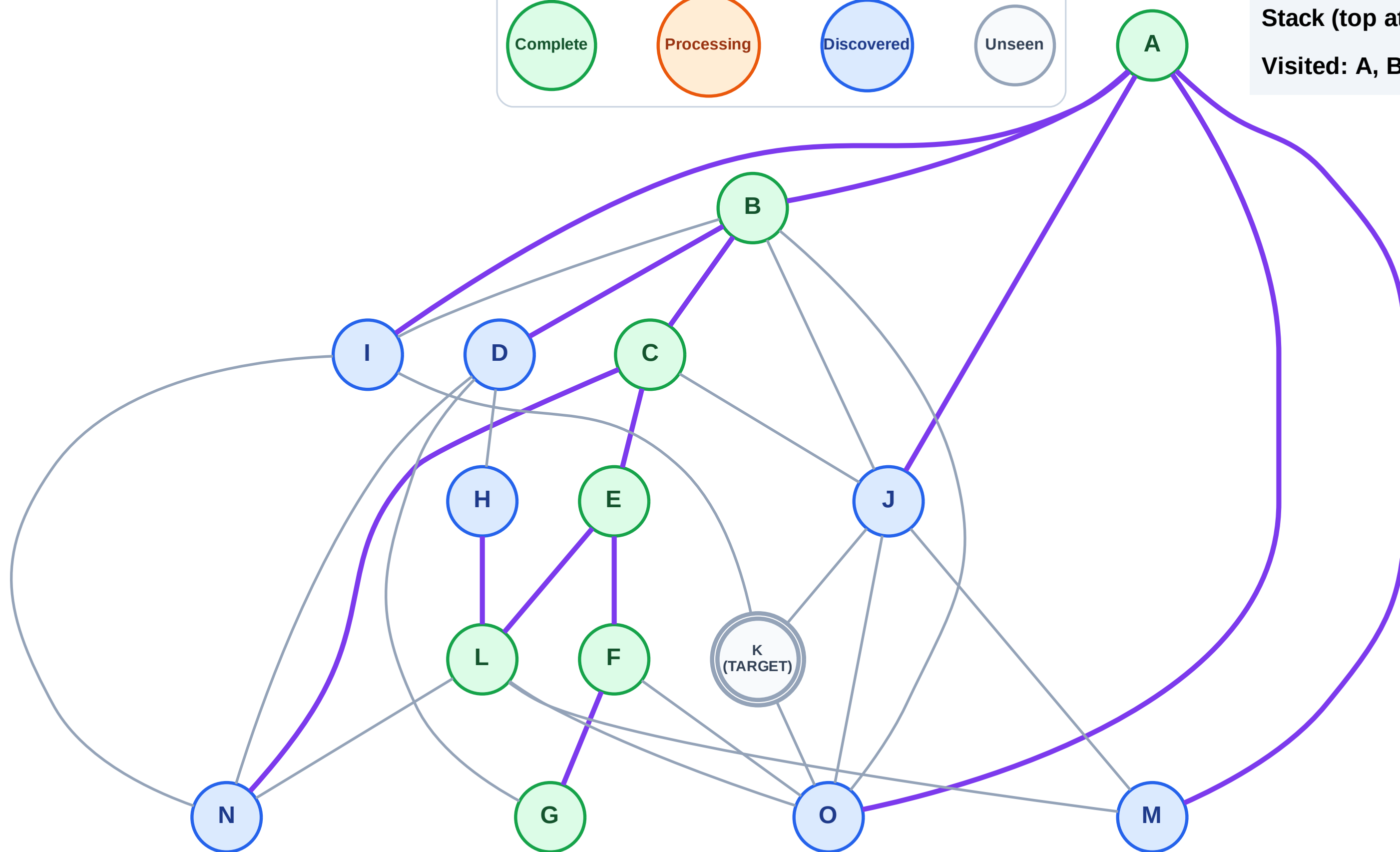
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N | H

Visited: A, B, C, E, F, G, L



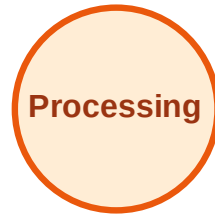
DFS Traversal

Step 16: Process node H

Legend



Complete



Processing



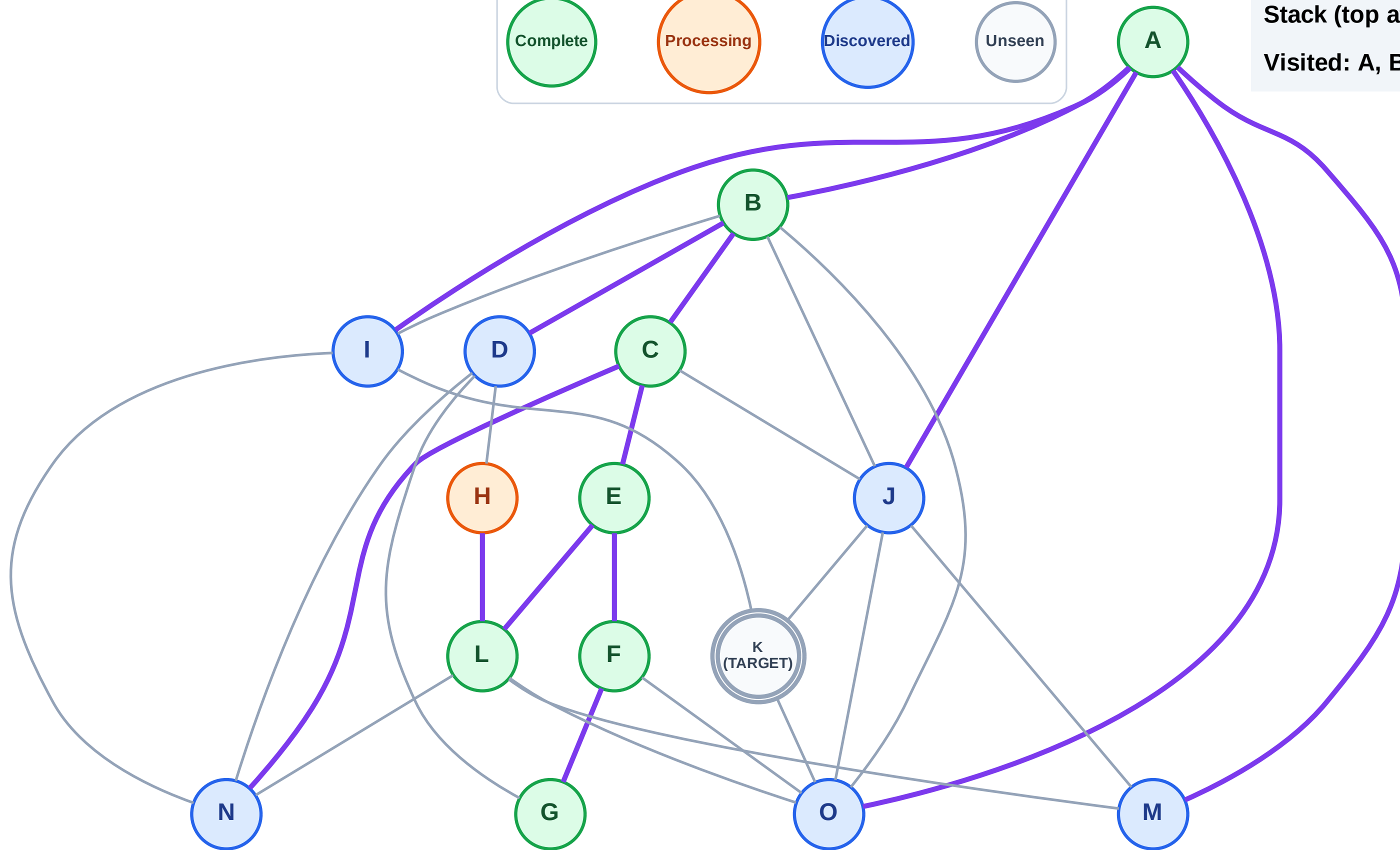
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N

Visited: A, B, C, E, F, G, L



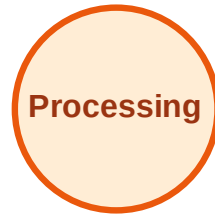
DFS Traversal

Step 17: No new neighbors from H

Legend



Complete



Processing



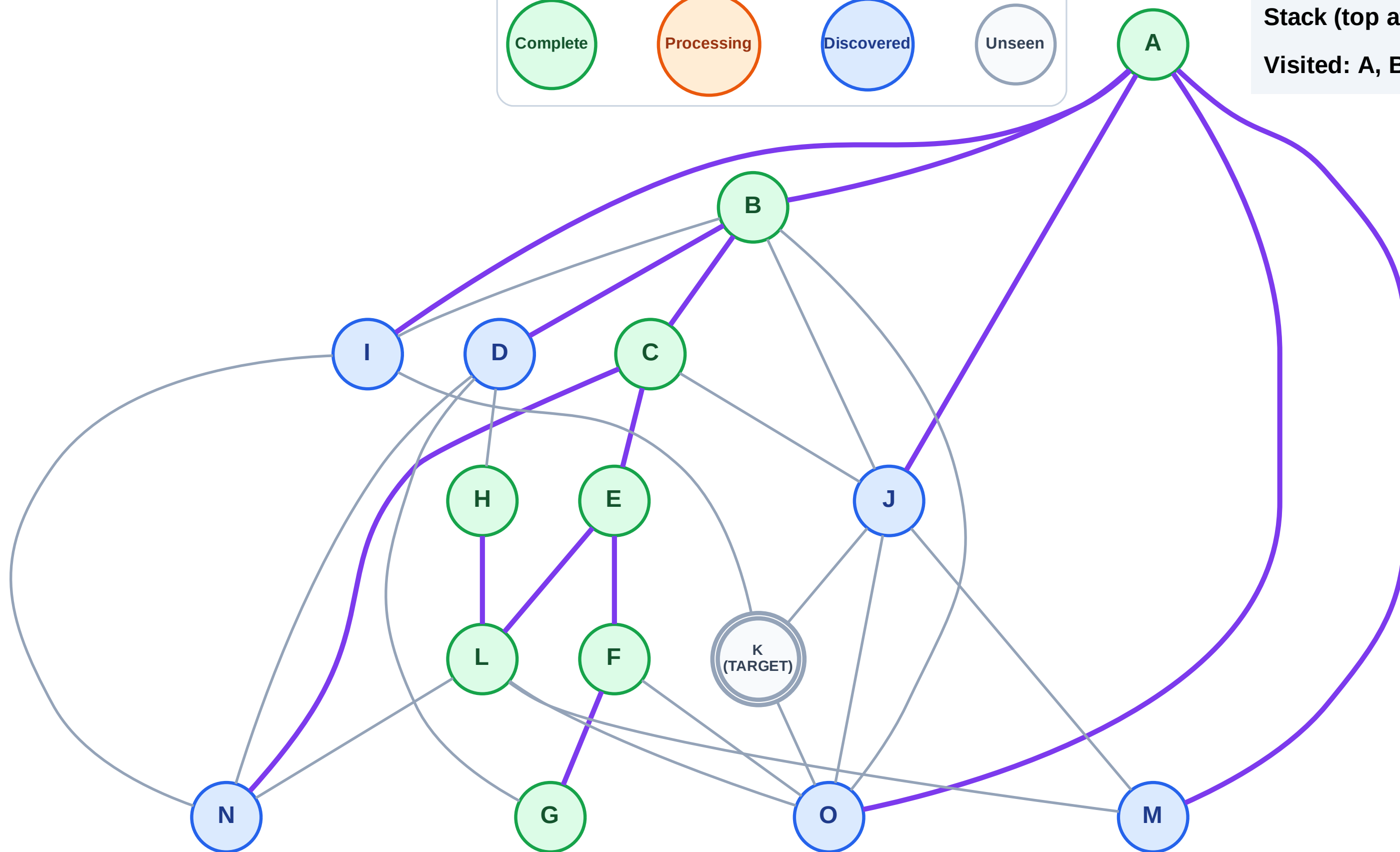
Discovered



Unseen

Stack (top at right): O | M | J | I | D | N

Visited: A, B, C, E, F, G, H, L



DFS Traversal

Step 18: Process node N

Legend

Complete

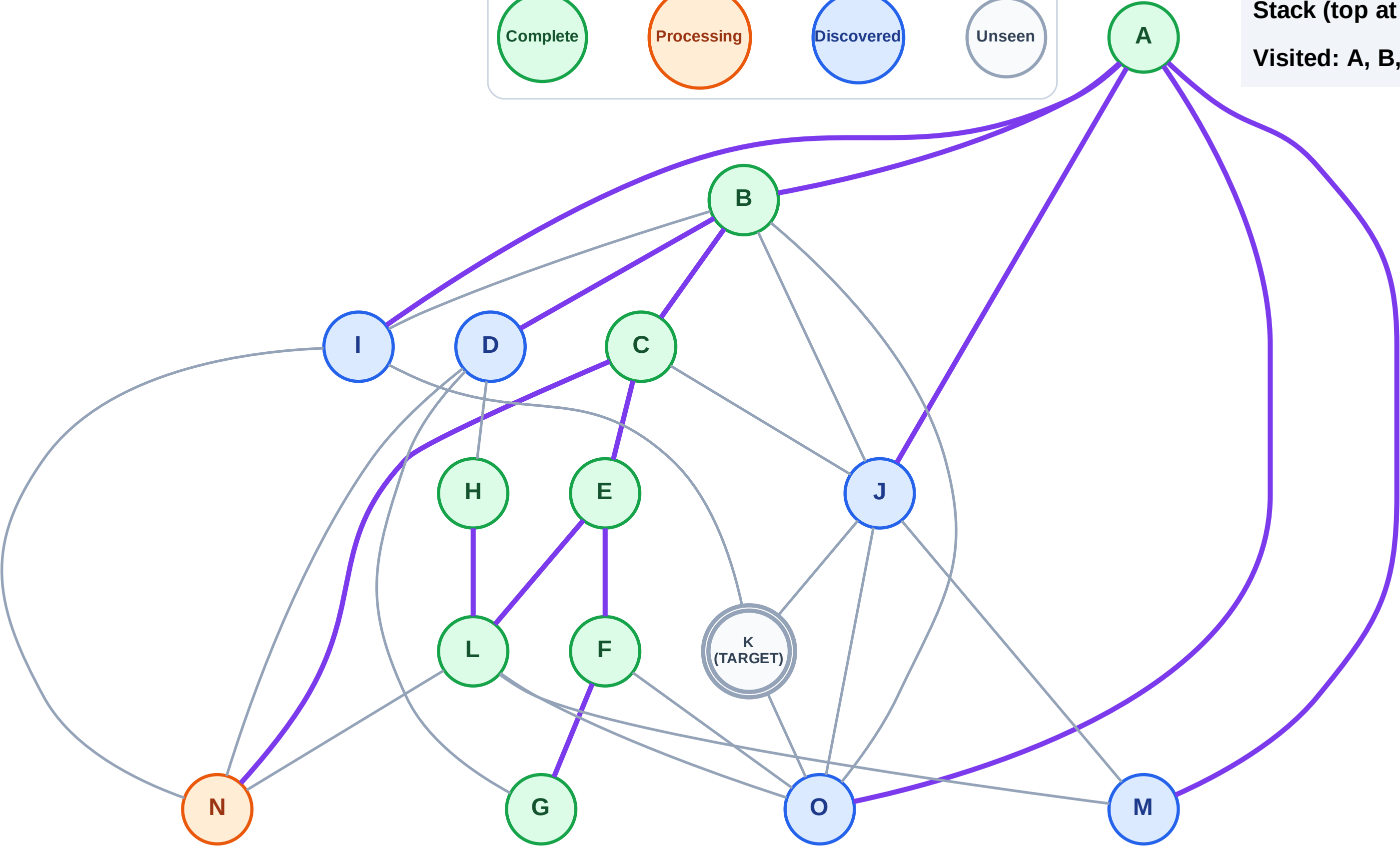
Processing

Discovered

Unseen

Stack (top at right): O | M | J | I | D

Visited: A, B, C, E, F, G, H, L



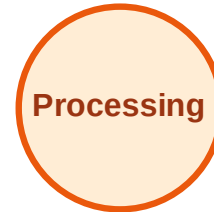
DFS Traversal

Step 19: No new neighbors from N

Legend



Complete



Processing



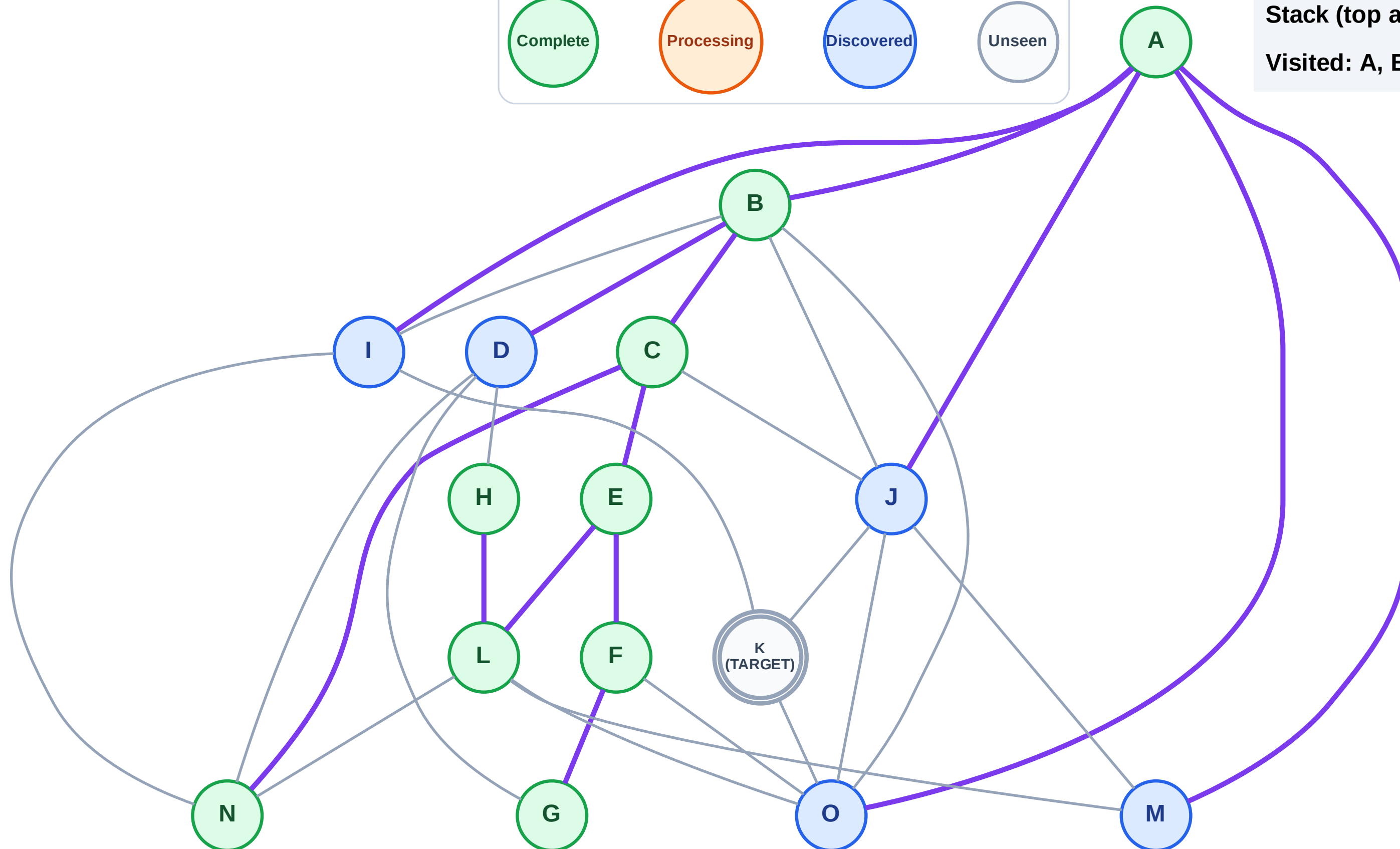
Discovered



Unseen

Stack (top at right): O | M | J | I | D

Visited: A, B, C, E, F, G, H, L, N



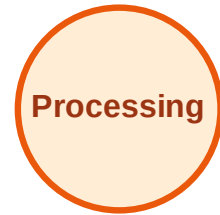
DFS Traversal

Step 20: Process node D

Legend



Complete



Processing



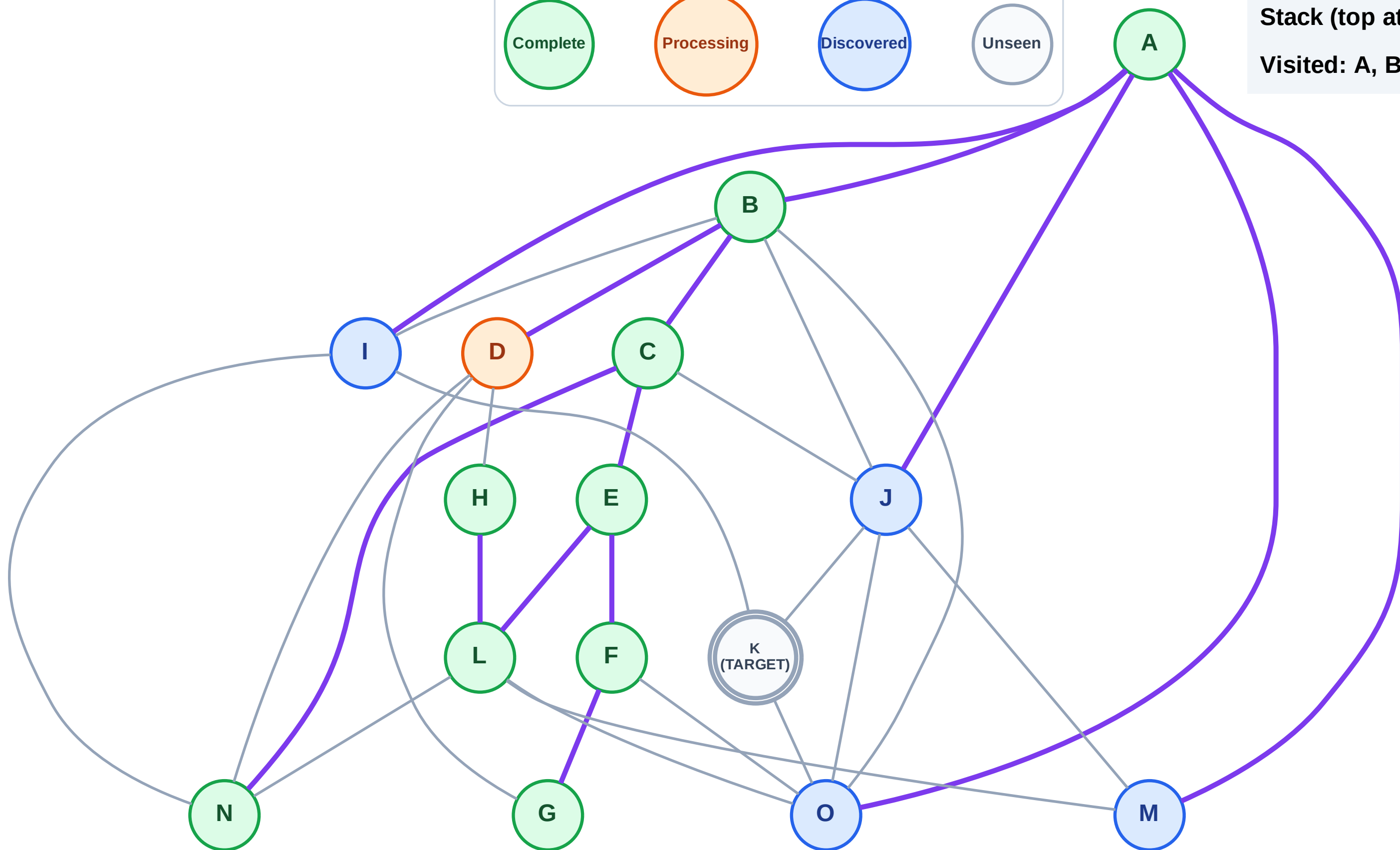
Discovered



Unseen

Stack (top at right): O | M | J | I

Visited: A, B, C, E, F, G, H, L, N



DFS Traversal

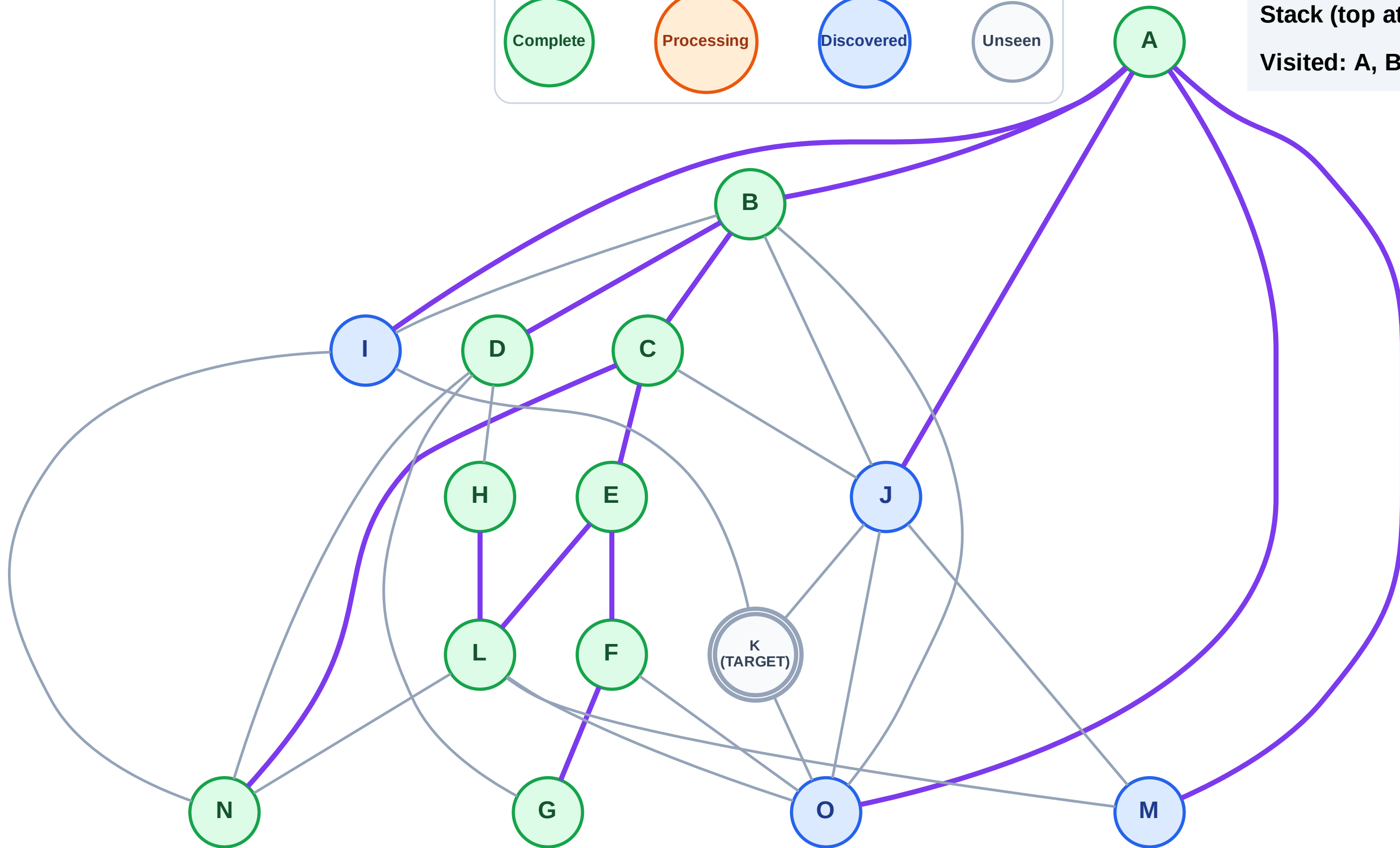
Step 21: No new neighbors from D

Legend



Stack (top at right): O | M | J | I

Visited: A, B, C, D, E, F, G, H, L, N



DFS Traversal

Step 22: Process node I

Legend

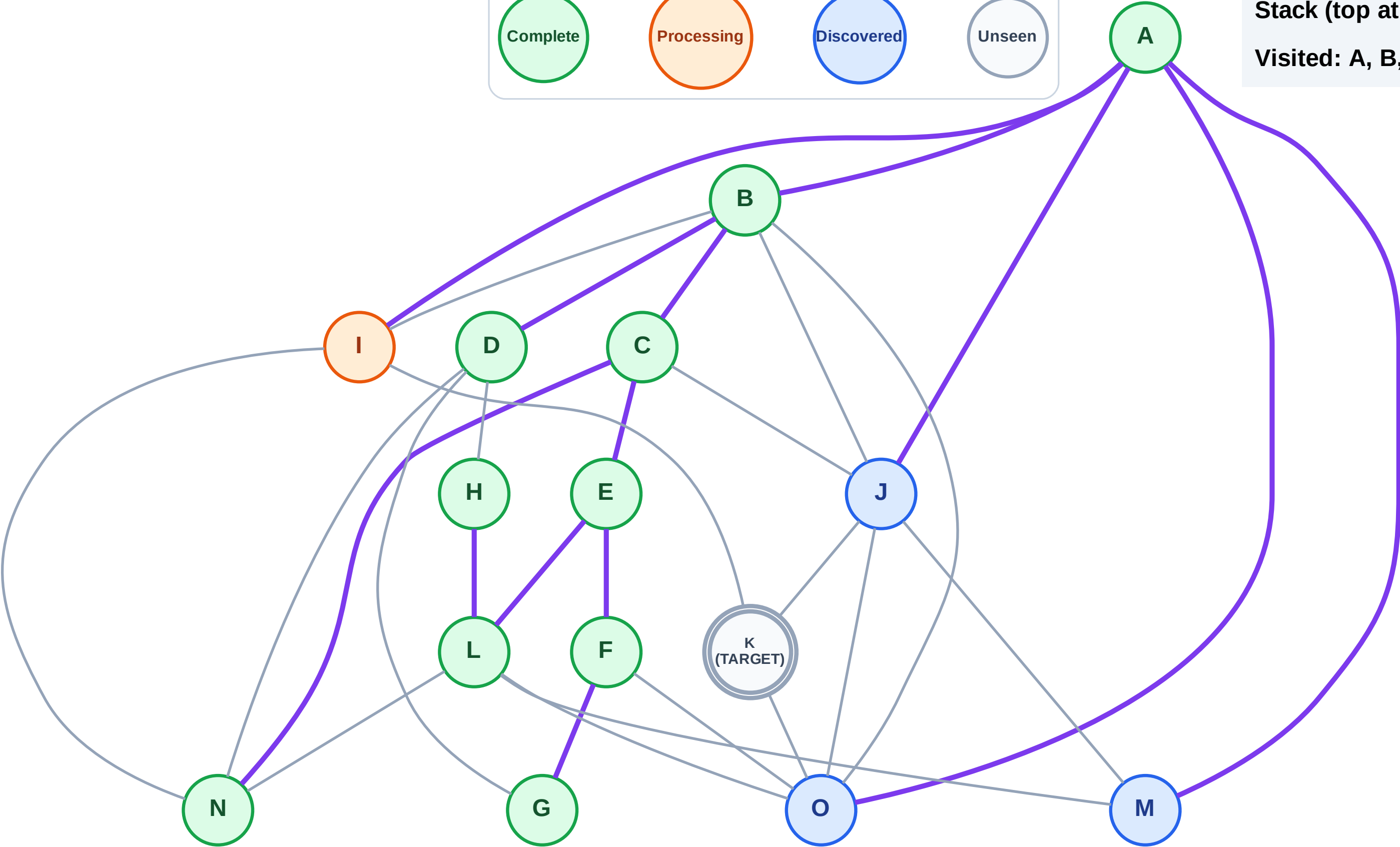
Complete

Processing

Discovered

Unseen

Stack (top at right): O | M | J
Visited: A, B, C, D, E, F, G, H, L, N



DFS Traversal

Step 23: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → I → K

Legend



Stack (top at right): O | M | J

Visited: A, B, C, D, E, F, G, H, I, K, L, N

